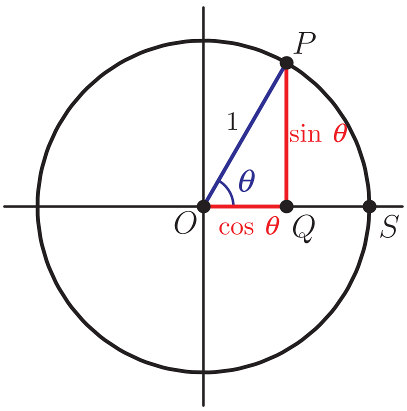
11A Right-angled triangles

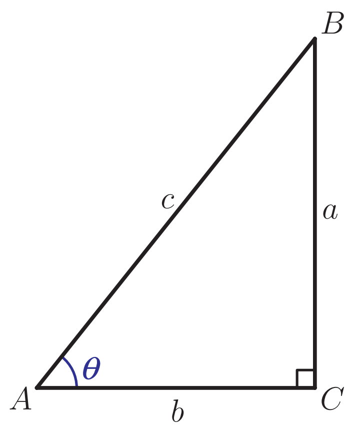
In previous studies, you may have already seen definitions of sine, cosine and tangent functions in terms of the sides of a right-angled triangle. In this section we will briefly discuss how they are related to the definitions we introduced in the previous chapter.

Remember the definition of sine and cosine functions using the unit circle. Let $0° < \theta < 90°$, and let P be the point on the unit circle such that SOP = θ. Then $PQ = \sin\theta$ and $OQ = \cos\theta$.

Trigonometric functions were defined in Section 9B.

Consider now a right-angled triangle ABC with right angle at C and $B\hat{A}C = \theta$.





Triangles OPQ and ABC have the same angles, so they are similar. Therefore:

$$\frac{a}{\sin\theta} = \frac{b}{\cos\theta} = \frac{c}{1}$$

From this we find that the ratios of sides in a right-angled triangle are trigonometric functions of angle θ.

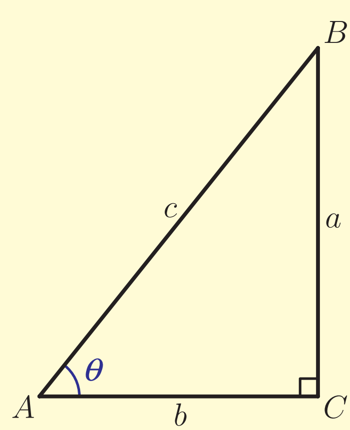
KEY POINT 11.1

In a right-angled triangle:

$$\frac{a}{c} = \sin\theta$$

$$\frac{b}{c} = \cos\theta$$

$$\frac{a}{b} = \frac{\sin\theta}{\cos\theta} = \tan\theta$$

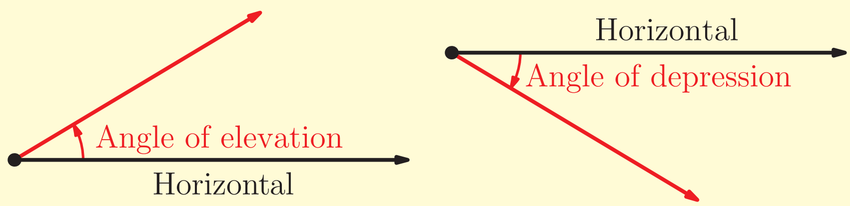


See Prior learning Section U on the CD-ROM for practice questions on right-angled triangles.

These equations only apply to right-angled triangles, so the angles are always acute. In the remainder of this chapter we shall deal with triangles which can also have obtuse angles. In those cases we need to use our definitions of trigonometric functions from the unit circle.

Two terms which you will see frequently in the context of trigonometry are **angle of elevation** and **angle of depression**.

KEY POINT 11.2



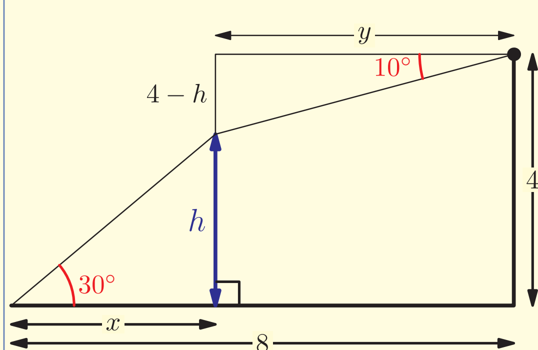
The angle of elevation is the angle above the horizontal.

The angle of depression is the angle below the horizontal.

Worked example 11.1

Daniel and Theo are trying to find the height of a birds' nest in their garden. From Theo's bedroom window, which is 4 m above the ground, the angle of depression of the nest is 10°. From the end of the flat garden, 8 m away from the house, the angle of elevation is 30°. Find the height of the nest.

Sketch a diagram



EXAM HINT

If a diagram is not given it is always a good use of time to sketch one.

Apply trigonometry to the right-angled triangles

$$y = \frac{4-h}{\tan 10°}$$

$$x = \frac{h}{\tan 30°}$$

continued . . .

From the diagram we can find an equation linking x and y

$$x + y = 8$$

$$\therefore \frac{4-h}{\tan 10°} + \frac{h}{\tan 30°} = 8$$

Evaluate tan 10 and tan 30 on the calculator

$$\Rightarrow \frac{4-h}{0.176} + \frac{h}{0.577} = 8$$

$$\Rightarrow 22.69 - 5.67h + 1.73h = 8$$

$$\Rightarrow 14.69 = 3.94h$$

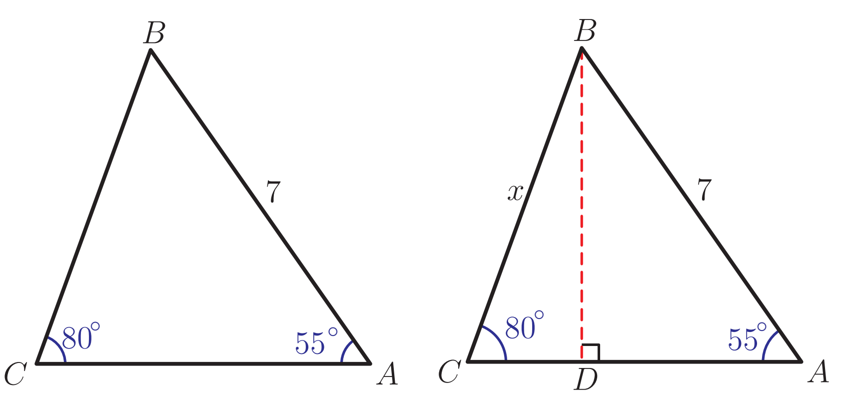
$$\Rightarrow h = 3.73\text{m (3SF)}$$

EXAM HINT

In a question with several steps like this, it is important not to round intermediate values. Use the answer button on your calculator and round at the end. However, you do not have to write down all the digits on your calculator display.

11B The sine rule

Can we use trigonometry to calculate sides and angles in triangles that do not have a right angle? The answer is yes, and one way to do this is called the **sine rule**. In the triangle shown on the left below, $AB = 7$, $B\hat{A}C = 55°$ and $A\hat{C}B = 80°$. Can we find the length of BC?



There are no right angles in the diagram, but we can create some by drawing the line $[BD]$ perpendicular to $[AC]$ as shown in the second diagram.

We now have two right-angled triangles. In triangle ABD,
$\frac{BD}{7} = \sin 55°$, so $BD = 7\sin 55°$.

In triangle BCD, $\frac{BD}{x} = \sin 80°$, so $BD = x\sin 80°$.

Comparing the two expressions for BD, we get:

$$x\sin 80° = 7\sin 55° \qquad (*)$$

And rearranging gives:

$$x = \frac{7\sin 55°}{\sin 80°} = 5.82$$

Notice that we did not actually need to calculate the length of BD but can go straight to equation marked (*). This equation is more commonly written as:

$$\frac{x}{\sin 55°} = \frac{7}{\sin 80°}$$

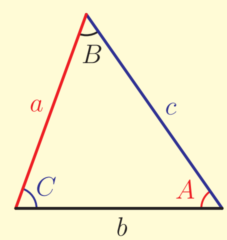
This last equation is an example of the sine rule. Notice that the length of each side is divided by the sine of the angle *opposite* that side. We can repeat the same process to obtain a general equation.

You may think that the sum of angles in a triangle being 180° is an absolute fact, but it is only the case on a flat plane. If you draw a triangle on a sphere, the sum of the angles will be greater than 180°. The study of these spherical triangles is vital to navigation and astronomy. There are even analogues of the cosine and sine rules for these triangles.

KEY POINT 11.3

The sine rule

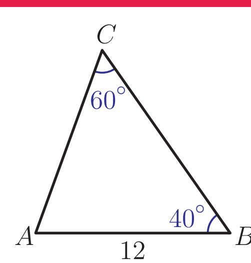
$$\frac{a}{\sin \hat{A}} = \frac{b}{\sin \hat{B}} = \frac{c}{\sin \hat{C}}$$



To use the sine rule you need to have both an angle and its opposite side. When using the sine rule you will normally use only two of the three ratios. To decide which ones, you need to look at what information is given in the question.

Worked example 11.2

Find the length of side AC.



continued . . .

We are given the angles opposite sides AB and AC, so use the sine rule with those two sides

$$\frac{12}{\sin 60^\circ} = \frac{AC}{\sin 40^\circ}$$

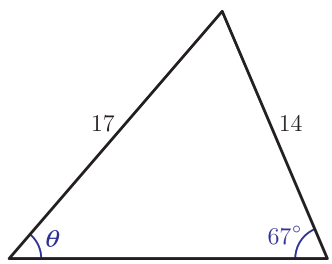
$$\Rightarrow AC = \frac{12\sin 40^\circ}{\sin 60^\circ}$$

$$= 8.91 \text{ (3SF)}$$

We can also use the sine rule to find angles.

Worked example 11.3

Find the size of the angle marked θ.



Use the sine rule with the two given sides, as we have one of the opposite angles and want to find the other one

$$\frac{17}{\sin 67^\circ} = \frac{14}{\sin\theta}$$

$$\Rightarrow \sin\theta = \frac{14\sin 67^\circ}{17} = 0.758$$

$$\therefore \theta = \arcsin 0.758 = 49.3^\circ$$

You should remember from your work on trigonometric equations that there is more than one value of θ with $\sin\theta = 0.758$. So does that mean that the last question has more than one possible answer?

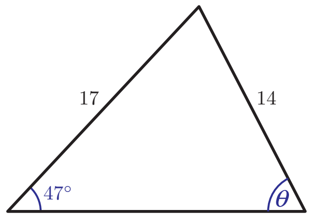
See Section 10A for solving trigonometric equations.

Another solution of the equation $\sin\theta = 0.758$ is $180 - 49.3 = 130.7^\circ$. However, as one of the other angles is 67°, this value is impossible, because all three angles of the triangle must add up to 180°, and $130.7 + 67 = 190.7 > 180$. All other possible values of θ are outside the interval [0°, 180°], so cannot be angles of a triangle. In this example, there is only one possible value of angle θ.

The next example shows that this is not always the case.

Worked example 11.4

Find the size of the angle marked θ, giving your answer to the nearest degree.



Use the sine rule with the two given sides

$$\frac{17}{\sin\theta} = \frac{14}{\sin 47°}$$

$$\Rightarrow \sin\theta = \frac{17\sin 47°}{14} = 0.888$$

Find the two possible values of θ

$\arcsin 0.888 = 62.6°$

$\Rightarrow \theta = 62.6°$ or $180 - 62.6 = 117.4°$

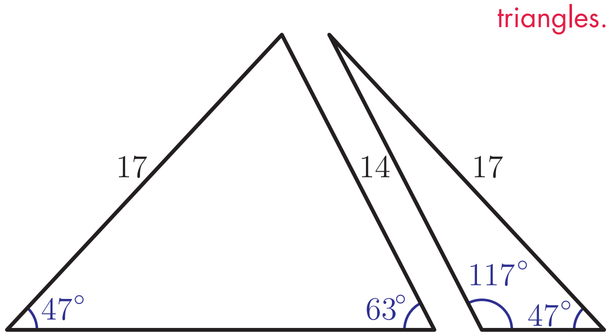
Check whether both solutions are possible: do the two angles add up to less than 180°?

$62.6 + 47 = 109.6 < 180$

$117.4 + 47 = 164.4 < 180$

Both solutions are possible.
The diagram below shows the two possible triangles.

$\therefore \theta = 63°$ or $117°$



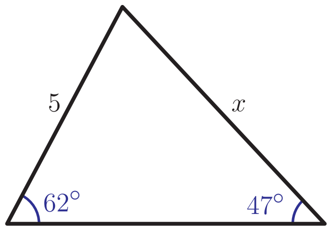
EXAM HINT

In an examination, a question will often alert you to look for two possible answers. However, if it doesn't, you should check whether the second solution is possible by finding the sum of the known angles.

Exercise 11B

1. Find the lengths of sides marked with letters.

(a) (i)



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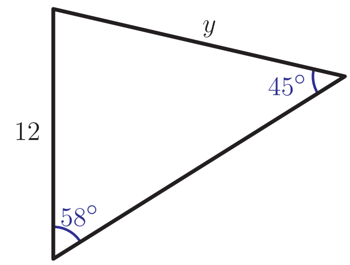
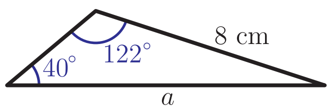


(b) (i)



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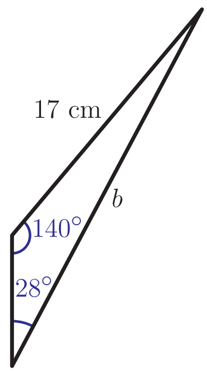
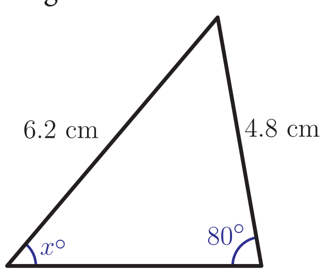
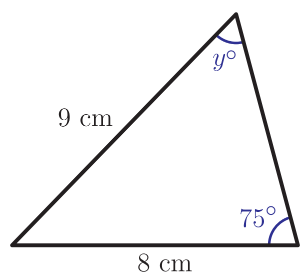


2. Find the angles marked with letters, checking whether there is more than one solution.

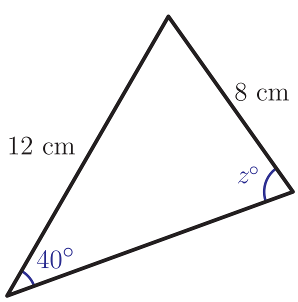
(a) (i)



(ii)



(b) (i)



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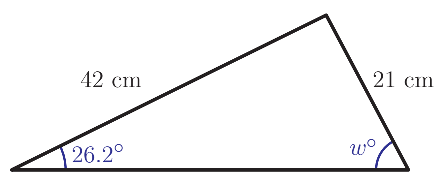
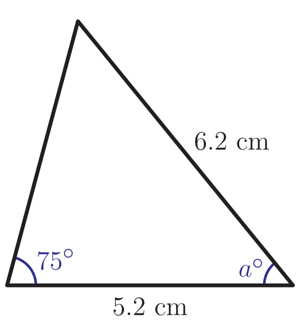
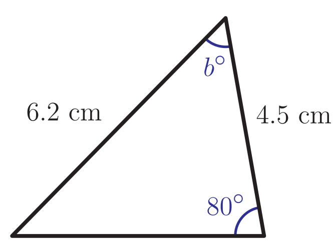


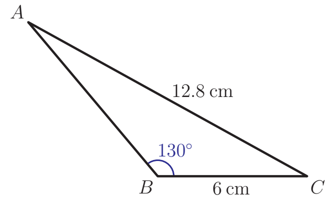
(c) (i)



(ii)

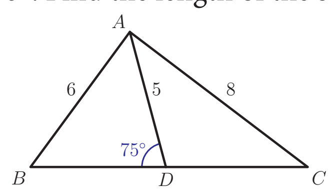


3. Find all the unknown sides and angles of triangle ABC.



4. In triangle ABC, $AB = 6\,\text{cm}$, $BC = 8\,\text{cm}$, $A\hat{C}B = 35°$. Show that there are two possible triangles with these measurements and find the remaining side and angles for each. *[4 marks]*

5. In the triangle shown in the diagram, $AB = 6$, $AC = 8$, $AD = 5$ and $A\hat{D}B = 75°$. Find the length of the side BC.



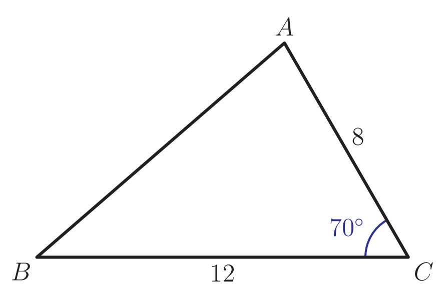
[5 marks]

6. Show that it is impossible to draw a triangle ABC with $AB = 12\,\text{cm}$, $AC = 8\,\text{cm}$ and $A\hat{B}C = 47°$. *[5 marks]*

11C The cosine rule

Not all information about triangles can be found using the sine rule. In particular, if we have two sides and the angle between them or all three sides we can find further information using a new rule called the **cosine rule**.

Can we find the length of the side AB in the triangle shown?



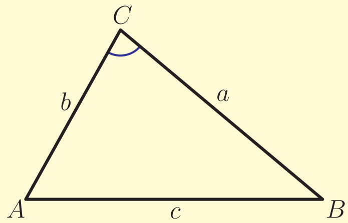
The sine rule for this triangle gives $\dfrac{AB}{\sin 70°} = \dfrac{8}{\sin \hat{B}} = \dfrac{12}{\sin \hat{C}}$. But we do not know either of the angles B or C, so it is impossible to find AB from this equation. We need a different strategy.

As before, by creating two right-angled triangles we can derive a formula for the length of AB (See Fill-in proof 8 'Cosine rule' for proof on the CD-ROM).

KEY POINT 11.4

The Cosine rule

$c^2 = a^2 + b^2 - 2ab\cos\hat{C}$



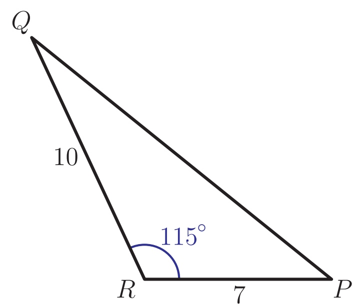
EXAM HINT

This is the form of the cosine rule given in the Formula booklet. However, you can change the names of the variables to anything you like as long as the angle corresponds to the side given on the left-hand side of the equation.

The cosine rule can be used to find the third side of the triangle when we know the other two sides and the angle between them.

Worked example 11.5

Find the length of the side PQ.



The question involves two sides and an angle, so use the cosine rule. The known angle is opposite PQ

$$PQ^2 = 7^2 + 10^2 - 2 \times 7 \times 10 \times \cos 115°$$

$$\Rightarrow PQ^2 = 208.2$$

$$\Rightarrow PQ = \sqrt{208.2} = 14.4$$

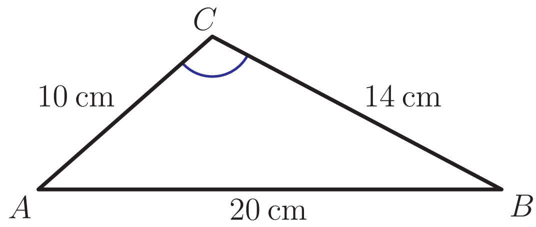
We can also use the cosine rule to find an angle if we know all three sides of a triangle. To help with this there is a rearrangement of the cosine rule:

KEY POINT 11.5

The cosine rule: $\cos\hat{C} = \dfrac{a^2 + b^2 - c^2}{2ab}$

Worked example 11.6

Find the size of the angle $A\hat{C}B$ correct to the nearest degree.



The question involves three sides and an angle, so use the cosine rule. Side AB is opposite the angle we want

$$\cos\hat{C} = \frac{196+100-400}{2\times14\times10}$$
$$= -\frac{104}{280}$$
$$= -0.371$$

Use inverse cosine to find the angle

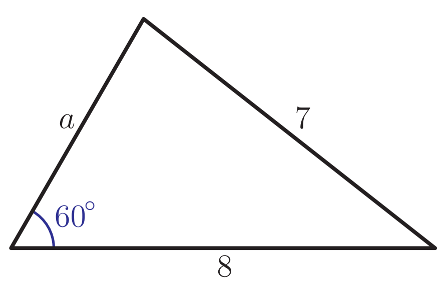
$$\hat{C} = \arccos(-0.371) = 112°$$

Notice that in the last two examples the angle is obtuse and its cosine is negative. Notice also that when using the cosine rule to find an angle, there is no second solution. This is because the second possibility is (360° – first solution) and so it will always be greater than 180°.

It is possible to use the cosine rule even when the given angle is not opposite the required side, as illustrated in the next example. This example also reminds you that there are some exact values of trigonometric functions you should know.

Worked example 11.7

Find the possible lengths of the side marked a.



continued . . .

The question involves three sides and an angle, so use the cosine rule. The known angle is opposite the side marked 7

$7^2 = a^2 + 8^2 - 2a \times 8\cos 60°$

Use the fact that $\cos 60° = \frac{1}{2}$

$\Rightarrow 49 = a^2 + 64 - 8a$

We recognise that this is a quadratic equation

$\Rightarrow a^2 - 8a + 15 = 0$

$\Rightarrow (a-3)(a-5) = 0$

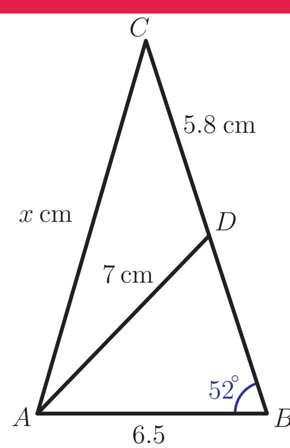
$\Rightarrow a = 3 \quad \text{or} \quad 5$

It is also possible to answer this question using the sine rule twice, first to find the angle opposite the side marked 8, and then to find side a. Try to see if you can get the same answers.

The next example illustrates how to select which of the two rules to use. For both sine and cosine rule, we need to know three measurements in a triangle to find a fourth one.

Worked example 11.8

In the triangle shown in the diagram, $AB = 6.5\,\text{cm}$, $AD = 7\,\text{cm}$, $CD = 5.8\,\text{cm}$, $A\hat{B}C = 52°$ and $AC = x$. Find the value of x correct to one decimal place.



The only triangle in which we know three measurements is ABD. We know two sides and a non-included angle (i.e. the angle does not lie between two given sides), so we can use the sine rule to find $A\hat{D}B$

Sine rule in triangle ABD; let $A\hat{D}B = \theta$:

$\frac{6.5}{\sin\theta} = \frac{7}{\sin 52°}$

$\Rightarrow \sin\theta = \frac{6.5\sin 52°}{7} = 0.7317$

$\arcsin 0.7317 = 47°$

continued . . .

Are there two possible solutions?

$180 - 47 = 133,\ 133 + 52 > 180$

There is only one solution.

$\therefore \theta = 47^\circ$ so, $A\hat{D}B = 47^\circ$

In triangle ADC, we know two sides and want to find the third. We can find $A\hat{D}C$ and then use the cosine rule.

$A\hat{D}C = 180 - 47 = 133^\circ$

Cosine rule in triangle ADC:

$x^2 = 7^2 + 5.8^2 - 2 \times 7 \times 5.8 \cos 133^\circ$

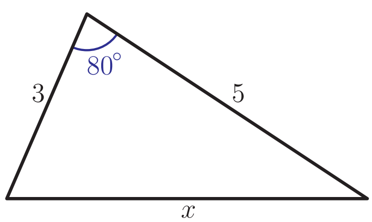
$\Rightarrow x^2 = 137.99$

$\Rightarrow x = \sqrt{137.99} = 11.7\text{cm}$

Exercise 11C

1. Find the lengths of the sides marked with letters.

(a) (i)



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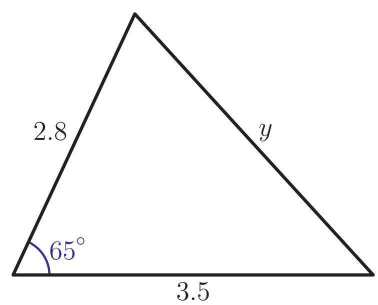
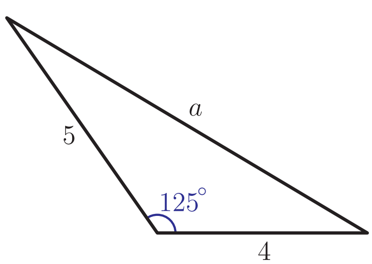


(b) (i)



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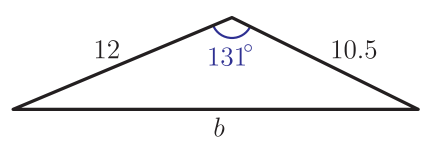
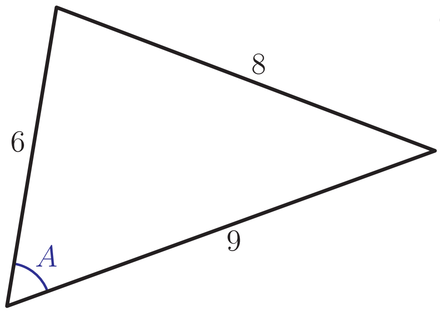
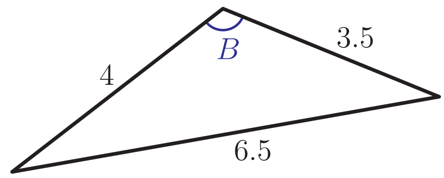


2. Find the angles marked with letters.

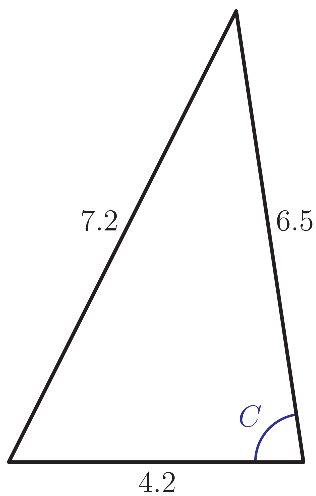
(a) (i)



(ii)



(b) (i)



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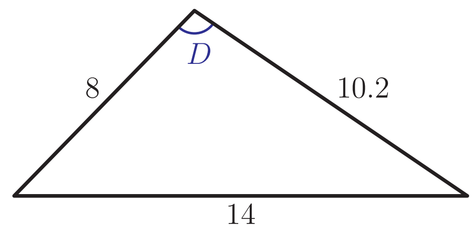
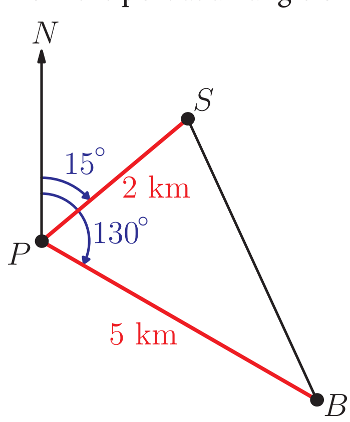


3. (i) Triangle PQR has sides $PQ = 8$ cm, $QR = 12$ cm, $RP = 7$ cm. Find the size of the largest angle.

(ii) Triangle ABC has sides $AB = 4.5$ cm, $BC = 6.2$ cm, $CA = 3.7$ cm. Find the size of the smallest angle.

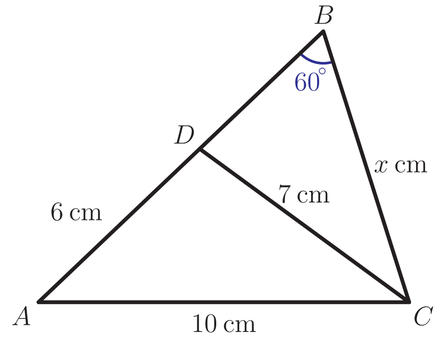
4. Ship S is 2 km from the port at an angle of 15° from North and boat B is 5 km from the port at an angle of 130° from North.



Find the distance between the ship and the boat. *[6 marks]*

Land surveying and astronomy were two areas which motivated the development of trigonometry. Supplementary sheet 4 'Coordinate systems and graphs' on the CD-ROM shows you some examples of how the sine and cosine rule can be applied in these areas.

5. Find the value of x in the diagram below.



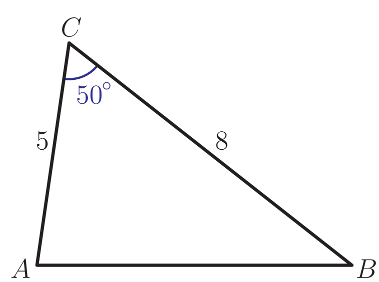
[6 marks]

6. In triangle ABC, $AB = (x - 3)$ cm, $BC = (x + 3)$ cm, $AC = 8$ cm and $B\hat{A}C = 60°$. Find the value of x. *[6 marks]*

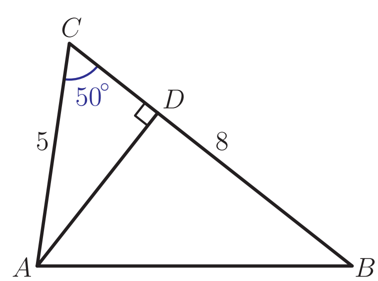
7. In triangle KLM, $KL = 4$, $LM = 7$ and $L\hat{K}M = 45°$. Find the exact length of KM. *[6 marks]*

11D Area of a triangle

Now that we know how to calculate sides and angles in a triangle, we can ask how to calculate the area. We know that the formula for the area of a triangle is $\frac{1}{2} \times \text{base} \times \text{perpendicular height}$ and we can use this to find the area of the triangle shown in the diagram:



We need to find a height of the triangle in order to calculate the area. For example, we can draw the line (AD) perpendicular to (BC), as in the second diagram:



Then $AD = 5\sin 50°$, so the area of the triangle is:

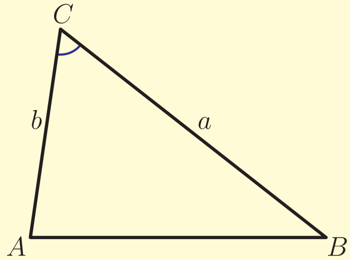
$$\frac{1}{2}BC \times AD = \frac{1}{2} \times 8 \times 5\sin 50° = 15.2\text{ cm}^2$$

This method can be applied to any triangle so the general formula for the area is:

KEY POINT 11.6

The area of the triangle is given by:

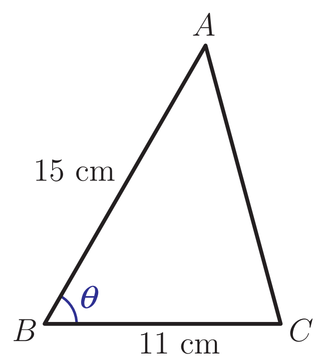
$$\text{area} = \frac{1}{2}ab\sin\hat{C}$$



There is also a formula for the area of a triangle if you know all three of its sides. It is called *Heron's formula*.

Worked example 11.9

The area of the triangle shown in the diagram is 52 cm². Find the two possible values of $A\hat{B}C$, correct to one decimal place.



Use the formula for the area of a triangle with known angle $\hat{B}$: area $= \frac{1}{2}ac\sin\hat{B}$

$$\frac{1}{2}(11 \times 15)\sin\theta = 52$$

$$\Rightarrow \sin\theta = \frac{2 \times 52}{11 \times 15} = 0.6303$$

$$\Rightarrow \arcsin 0.6303 = 39.07°$$

$$\therefore \theta = 39.1° \text{ or } 180 - 39.1 = 140.9°$$

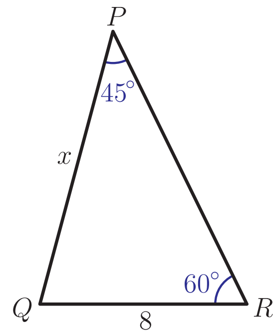
The next example combines the area of the triangle with the sine rule and working with exact values.

Worked example 11.10

For triangle PQR shown in the diagram:

(a) Calculate the exact value of x.

(b) Find the area of the triangle.



The question involves two angles and two sides, so we can use the sine rule

(a) $\frac{8}{\sin 45°} = \frac{x}{\sin 60°}$

We need the exact value of x, so use the exact values for $\sin 45°$ and $\sin 60°$

$\Rightarrow \frac{8}{\sqrt{2}/2} = \frac{x}{\sqrt{3}/2}$

$\Rightarrow \frac{16}{\sqrt{2}} = \frac{2x}{\sqrt{3}}$

$\Rightarrow x = \frac{16\sqrt{3}}{2\sqrt{2}}$

$\Rightarrow x = 4\sqrt{6}$

To use the formula for the area of the triangle, we need $P\hat{Q}R$

(b) $P\hat{Q}R = 180 - 60 - 45 = 75°$

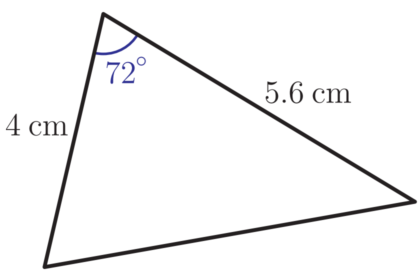
$\therefore \text{area} = \frac{1}{2}\left(8 \times 4\sqrt{6}\right)\sin 75°$

$= 37.9 \text{ (3SF)}$

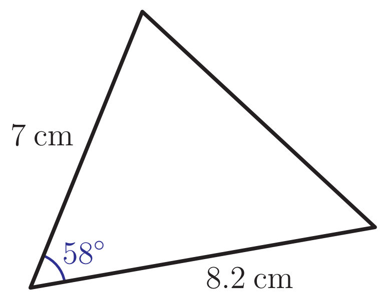
Exercise 11D

1. Calculate the areas of these triangles.

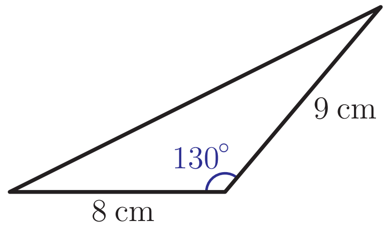
(a) (i)



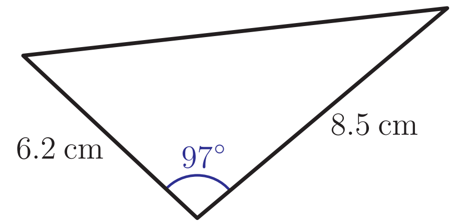
(ii)



(b) (i)

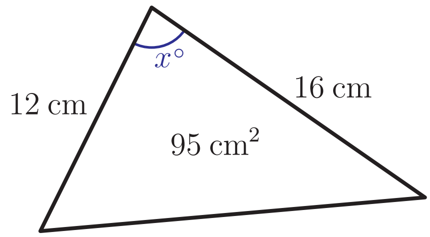


(ii)

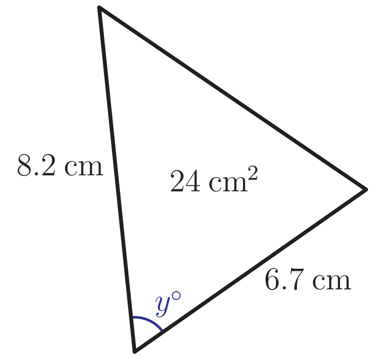


2. Each triangle has the area shown. Find two possible values of each marked angle.

(a)



(b)



3. In triangle LMN, $LM = 12$ cm, $MN = 7$ cm and $L\hat{M}N = 135°$.
Find the length of the side LN and the area of the triangle.

[6 marks]

4. In triangle PQR, $PQ = 8$ cm, $RQ = 7$ cm and $R\hat{P}Q = 60°$.
Find the exact difference in areas between the two possible triangles.

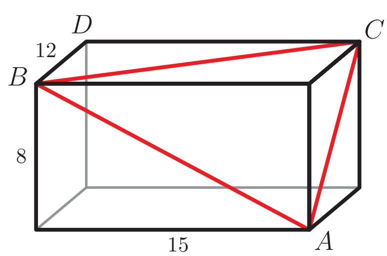
[6 marks]

11E Trigonometry in three dimensions

In many applications we work with three-dimensional objects. Examples in this section show you how to apply trigonometry in three dimensions. The general strategy is to identify a suitable triangle and then apply one of the rules from the previous sections.

Worked example 11.11

A cuboid has sides of length 8 cm, 12 cm and 15 cm. Diagonals of three of the faces are drawn as shown.



continued . . .

(a) Find the lengths of AB, BC and CA.

(b) Find the size of the angle $A\hat{C}B$.

(c) Calculate the area of the triangle ABC.

(d) Find the length AD.

AB is the diagonal of the front face, so it is a hypotenuse of a right-angled triangle with sides 8 and 12

(a) $AB^2 = 15^2 + 8^2 = 289$

$AB = \sqrt{298} = 17$ cm

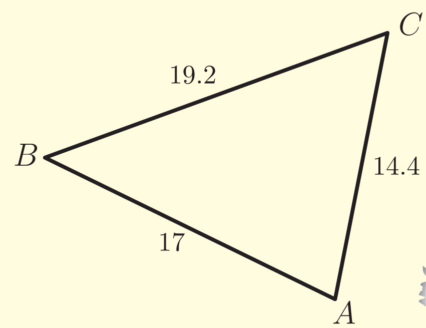
Find BC and CA in a similar way

$BC = \sqrt{12^2 + 15^2} = 19.2$ cm

$CA = \sqrt{12^2 + 8^2} = 14.4$ cm

We now look at triangle ABC. Draw the triangle if it helps

(b)



We know all three sides and want to find the angle, so we use the cosine rule

$$\cos\hat{C} = \frac{14.4^2 + 19.2^2 - 17^2}{2 \times 14.4 \times 19.2} = 0.519$$

$$C = \arccos 0.519 = 58.7°$$

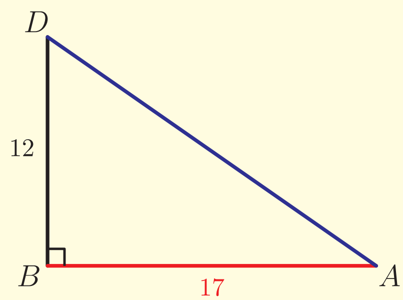
Use the formula for the area, with the angle we have just found

(c) $\text{area} = \frac{1}{2}(14.4 \times 19.2)\sin 58.7°$

$= 118$ cm^2

ABD is a right-angled triangle

(d)



$AD^2 = 12^2 + 17^2$

$AD = \sqrt{433} = 20.8$ cm

Part (d) illustrates a very useful fact about the diagonal of a cuboid.

KEY POINT 11.7

The diagonal of a $p \times q \times r$ cuboid has length $\sqrt{p^2 + q^2 + r^2}$.

In chapter 13 you will meet vectors which can also be used to solve three-dimensional problems.

The key to solving three-dimensional problems is spotting right angles. This is not always easy, as diagrams are drawn in perspective, but there are some common configurations to look for, such cross-sections and bisectors of isosceles triangles.

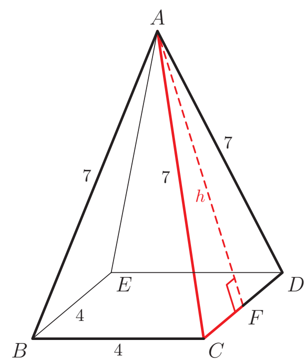
Worked example 11.12

The base of a pyramid is a square of side 4 cm. The other four faces are isosceles triangles with sides 7 cm. The height of one of the side faces is labelled h.

(a) Find the exact value of h.

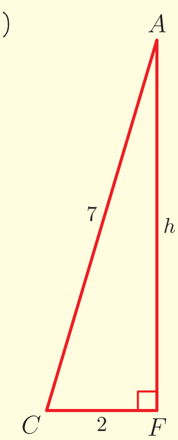
(b) Find the exact height of the pyramid.

(c) Calculate the volume of the pyramid, correct to 3 significant figures.



Triangle AFC is right angled. Draw it separately and label known sides

(a)



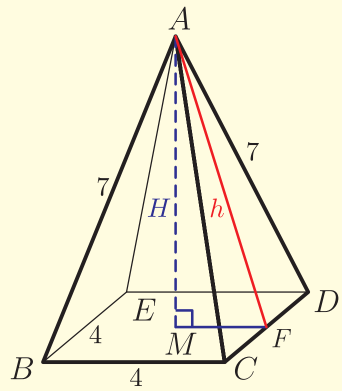
Use Pythagoras to find h

$h^2 = 7^2 - 2^2$
$h = \sqrt{45}$ cm

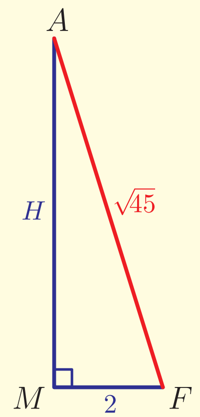
continued . . .

Add the height to the diagram. It is perpendicular to the base, so it makes a right-angle with MF. M is in the centre of the base, so $MF = 2$ cm

(b)



Look at triangle AMF



Use Pythagoras to find H

$H^2 = (\sqrt{45})^2 - 2^2$

$H = \sqrt{41}$ cm

From the Formula booklet, the formula for the volume of a pyramid is $\frac{1}{3}$(area of the base × height)

(c) $V = \frac{1}{3}(4^2 \times \sqrt{41})$

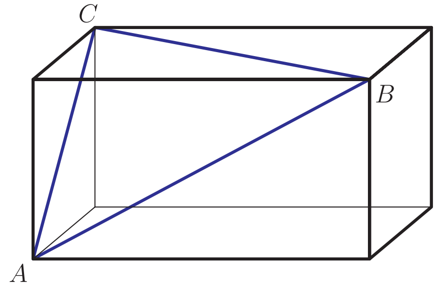
$V = 34.1\text{ cm}^3$

See Prior learning Section V on the CD-ROM for volumes of three-dimensional shapes.

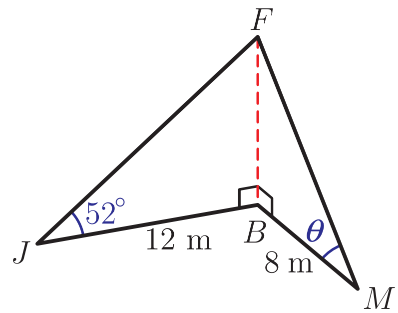
Exercise 11E

1. Find the length of the diagonal of the following cuboids:
   (a) 3 cm × 5 cm × 10 cm (b) 4 cm × 4 cm × 8 cm

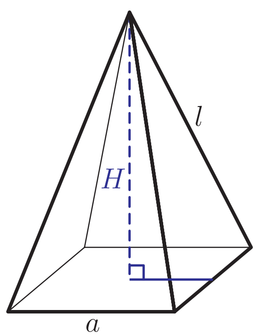
2. A cuboid has sides 12.5 cm, 10 cm and 7.3 cm. It is intersected by a plane passing through vertices A, B and C. Find the angles and the area of triangle ABC. *[8 marks]*



3. Johann stands 12 m from the base of a flagpole and sees the top at the angle of elevation of 52°. Marit stands 8 m from the flagpole. At what angle of elevation does she see the top? *[6 marks]*



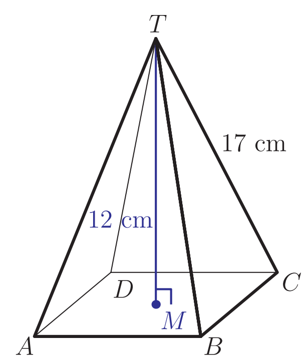
4. A square-based pyramid has a base of side $a = 8$ cm and a height $H = 12$ cm. Find the length of the sloping side, l. *[6 marks]*



5. The base of a pyramid $TABCD$ is a square. The height of the pyramid $TM = 12$ cm and the length of a sloping edge $TC = 17$ cm.

(a) Calculate the length of MC.

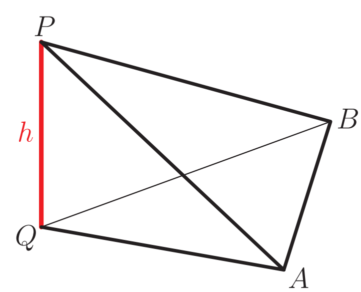
(b) Find the length of a side of the base. *[6 marks]*



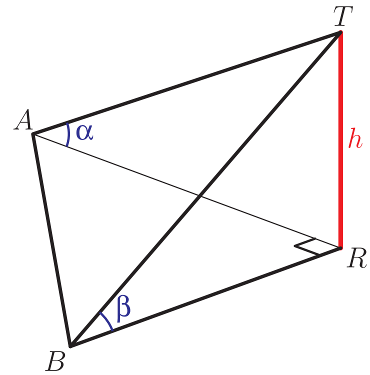
6. The diagram shows a vertical tree PQ and two observers at A and B, standing on horizontal ground. $AQ = 25$ m, $Q\hat{A}P = 37°$, $Q\hat{B}P = 42°$ and $A\hat{Q}B = 75°$.

(a) Find the height of the tree, h.

(b) Find the distance between the two observers. *[8 marks]*



7. Annabelle and Berta are trying to measure the height, h, of a vertical tree RT. They stand on horizontal ground, a distance d apart so that ARB is a right-angle. From where Annabelle is standing, the angle of elevation of the top of the tree is α, and from where Berta stands the angle of elevation of the top of the tree is β.

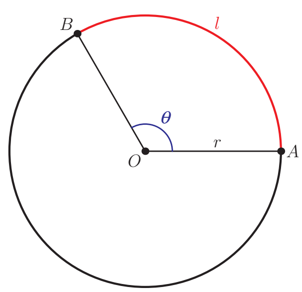


(a) Find expressions for RA and RB in terms of h, α and β, and hence show that:

$$h^2\left(\frac{1}{\tan^2\alpha}+\frac{1}{\tan^2\beta}\right)=d^2$$

(b) Given that $d = 26$ m, $\alpha = 45°$ and $\beta = 30°$, find the height of the tree. *[8 marks]*

11F Length of an arc



The diagram shows a circle with centre O and radius r, and points A and B on its circumference. The part of the circumference between points A and B is called an **arc** of a circle. You can see that there are in fact two such parts; the shorter one is called the **minor arc,** and the longer one the **major arc.** We say that the minor arc AB **subtends** angle θ at the centre of the circle.

You know that a measure of angle θ in radians is equal to the ratio of the length of the arc AB, l, to the circumference of the circle; in other words, $\theta = \frac{l}{r}$.

Radian measure was introduced in Section 9A.

This gives us a very simple formula to calculate the length of an arc of a circle if we know the angle it subtends at the centre.

KEY POINT 11.8

Length of an arc of a circle

$l = \theta r$

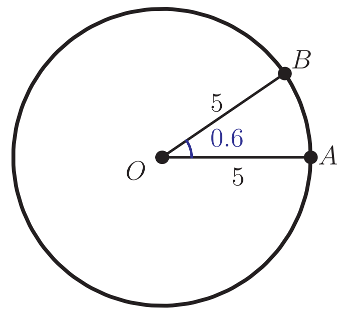
where r is the radius of the circle and θ is the angle subtended at the centre, measured in radians.

Worked example 11.13

Arc AB of a circle with radius 5 cm subtends an angle of 0.6 at the centre, as shown on the diagram.

(a) Find the length of the minor arc AB.

(b) Find the length of the major arc AB.



We know the formula for the length of an arc

(a) $l = r\theta$

$l = 5 \times 0.6$

$= 3\text{ cm}$

The angle subtended by the major arc is equal to a full turn minus the smaller angle. A full turn is 2π radians

(b) $\theta_1 = 2\pi - 0.6$

$= 5.683$

$l = r\theta_1 = 5 \times 5.683$

$= 28.4\text{ cm}\,(3\text{SF})$

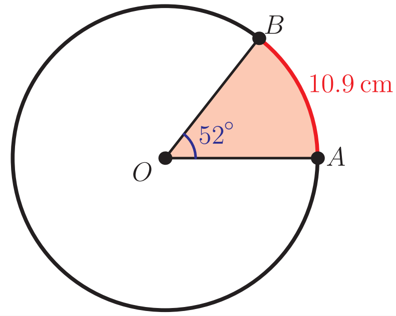
We could also have done the second part differently, by finding the length of the whole circumference and then taking away the minor arc: circle circumference is $2\pi r = 2\pi \times 5 = 31.42$, so the length of the major arc is $31.42 - 3 = 28.4\text{ cm}\,(3\text{ SF})$.

If the angle is given in degrees, we need to convert to radians before using the formula for the arc length.

Worked example 11.14

Two points, A and B, lie on the circumference of a circle of radius r cm. The arc AB has length 10.9 cm and subtends an angle of 52° at the centre of the circle.

(a) Find the value of r.
(b) Calculate the perimeter of the shaded region.



We know the arc length and the angle, so we can find the radius

(a) $l = r\theta \Rightarrow r = \dfrac{l}{\theta}$

To use the formula $l = r\theta$, the angle must be in radians

$\theta = 52 \times \dfrac{\pi}{180} = 0.908$

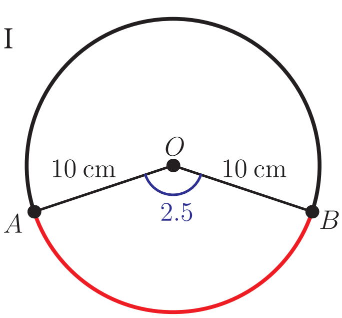
$\therefore r = \dfrac{10.9}{0.908} = 12.0\,\text{cm}$

The perimeter is made up of two radii and the arc

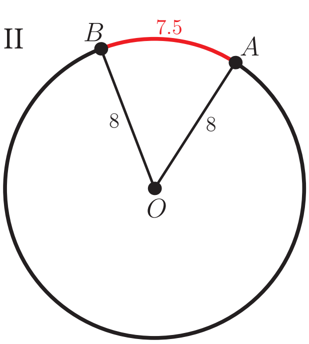
(b) $p = 2r + l$
$= 2 \times 12.0 + 10.9$
$= 34.9\,\text{cm}$

Exercise 11F

I



II



1. Calculate the length of the minor arc subtending an angle of θ radians at the centre of a circle of radius r cm.

(a) $\theta = 1.2, \quad r = 6.5$ (b) $\theta = 0.4, \quad r = 4.5$

2. Points A and B lie on the circumference of a circle with centre O and radius r cm.

Angle $A\hat{O}B$ is θ radians. Calculate the length of the major arc AB.

(a) $r = 15, \quad \theta = 0.8$ (b) $r = 1.4, \quad \theta = 1.4$

3. Calculate the length of the minor arc AB in the upper diagram (I). *[4 marks]*

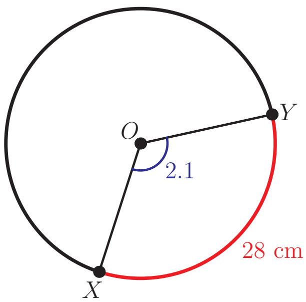
4. In the lower diagram (II), the radius of the circle is 8 cm and the length of the minor arc AB is 7.5 cm. Calculate the size of the angle $A\hat{O}B$:

(a) in radians
(b) in degrees. *[5 marks]*

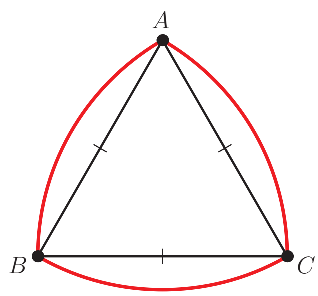
5. Points M and N lie on the circumference of a circle with centre C and radius 4 cm. The length of the *major* arc MN is 15 cm. Calculate the size of the *smaller* angle $M\hat{C}N$. *[4 marks]*

6. Points P and Q lie on the circumference of a circle with centre O. The length of the minor arc PQ is 12 cm and $P\hat{O}Q = 1.6$. Find the radius of the circle. *[4 marks]*

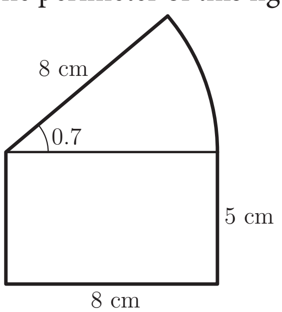
7. In the diagram, the length of the major arc XY is 28 cm. Find the radius of the circle. *[4 marks]*



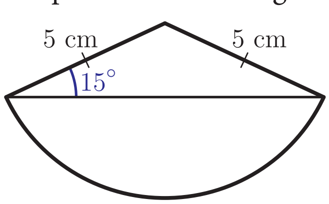
8. The figure below shows an equilateral triangle ABC with side $a = 5$ cm, and three arcs of circles with centres at the vertices of the triangle. Calculate the perimeter of the figure. *[5 marks]*



9. Calculate the perimeter of this figure: *[6 marks]*

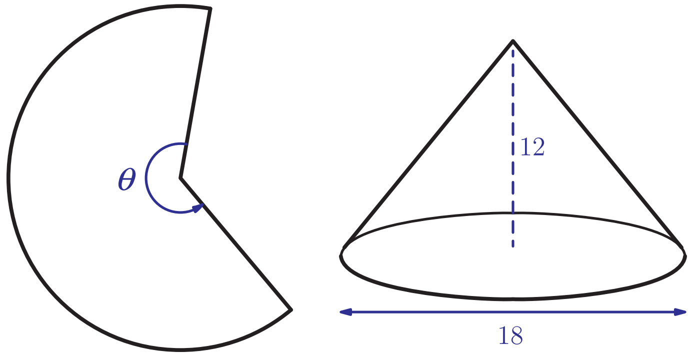


10. Find the exact perimeter of the figure shown: *[6 marks]*



11. A sector of a circle has perimeter $p = 12$ cm and angle at the centre $\theta = 0.4$. Find the radius of the circle. *[5 marks]*

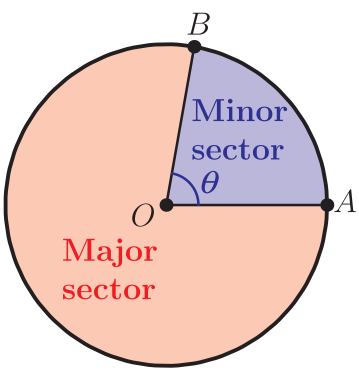
12. A cone is made by rolling a piece of paper as shown in the diagram.



If the cone is to have height 12 cm and base diameter 18 cm, find the size of the angle marked θ. *[6 marks]*

11G Area of a sector

A sector is a part of a circle bounded by two radii. As with arcs, we distinguish between a **minor sector** and a **major sector**.



The ratio of the area of the sector to the area of the whole circle is the same as the ratio of angle θ to a full turn. If A is the area of the sector, and angle θ is measured in radians, this means that $\frac{A}{\pi r^2} = \frac{\theta}{2\pi}$. Rearranging this equation gives the formula for the area of the sector.

KEY POINT 11.9

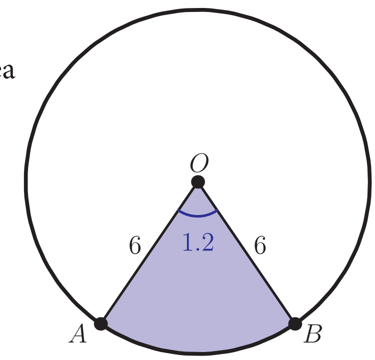
The area of a sector of a circle

$$A = \frac{1}{2} r^2 \theta,$$

where r is the radius of the circle and θ is the angle subtended at the centre, measured in radians.

Worked example 11.15

The diagram shows a circle with centre O and radius 6 cm, and two points on its circumference such that $A\hat{O}B = 1.2$. Find the area of the minor sector AOB.



Formula for the area of a sector

$$A = \frac{1}{2}r^2\theta$$

$$A = \frac{1}{2} \times 6^2 \times 1.2 = 21.6 \text{ cm}^2$$

We may also have to use the formulae for area and arc length in reverse.

Worked example 11.16

A sector of a circle has perimeter $p = 12$ cm and angle at the centre $\theta = 50°$.

Find the area of the sector.

If we are going to use the formula for the area of a sector, $A = \frac{1}{2}r^2\theta$, we first need to find r

We are given the perimeter, and we know that it consists of arc length $l = r\theta$ and two radii

$$p = r\theta + 2r$$

The angle needs to be in radians

$$\theta = 50 \times \frac{\pi}{180} = 0.873$$

Substitute all the values into the formula for perimeter to find r

$$12 = 0.873r + 2r = 2.873r$$

$$r = \frac{12}{2.873} = 4.177$$

Substitute θ and r into the formula for sector area

$$A = \frac{1}{2}(4.177)^2(0.873)$$

$$A = 7.62 \text{ cm}^2 \text{ (3SF)}$$

Exercise 11G

1. Points M and N lie on the circumference of a circle, centre O and radius r cm, with $M\hat{O}N = \alpha$. Calculate the area of the minor sector MON.

   (a) $r = 5, \alpha = 1.3$ (b) $r = 0.4, \alpha = 0.9$

2. Points A and B lie on the circumference of a circle with centre C and radius r cm. The size of the angle ACB is θ radians. Calculate the area of the major sector $A\hat{C}B$.

   (a) $r = 13, \quad \theta = 0.8$ (b) $r = 1.4, \quad \theta = 1.4$

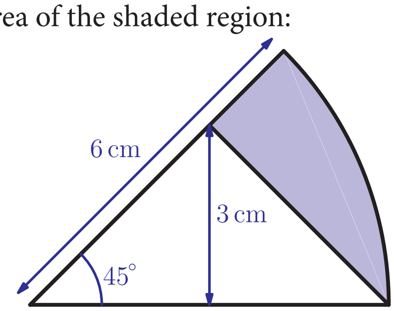
3. A circle has centre O and radius 10 cm. Points A and B lie on the circumference of the circle so that the area of the minor sector AOB is 40 cm^2. Calculate the size of the smaller angle $A\hat{O}B$. *[5 marks]*

4. Points P and Q lie on the circumference of a circle with radius 21 cm. The area of the *major* sector POQ is 744 cm^2. Find the size of the *smaller* angle $P\hat{O}Q$ in *degrees*. *[5 marks]*

5. A sector of a circle with angle 1.2 radians has area 54 cm^2. Find the radius of the circle. *[4 marks]*

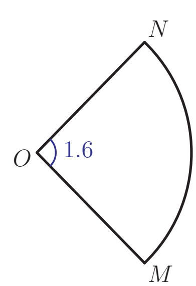
6. A sector of a circle with angle 162° has area 180 cm^2. Find the radius of the circle. *[4 marks]*

7. Find the area of the shaded region:



[6 marks]

8. The perimeter of the sector shown in the diagram is 28 cm. Find its area.



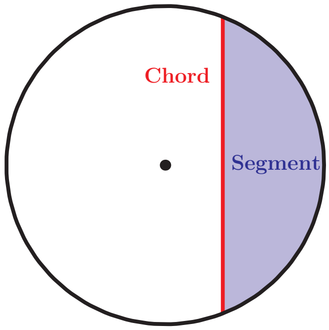
[5 marks]

9. A sector of a circle has perimeter 7 cm and area 3 cm². Find the possible values of the radius of the circle. *[6 marks]*

10. Points P and Q lie on the circumference of the circle with centre O and radius 5 cm. The difference between the areas of the major sector POQ and the minor sector POQ is 15 cm². Find the size of the angle $P\hat{O}Q$. *[5 marks]*

11H Triangles and circles

You already know how to calculate lengths of arcs and areas of sectors of circles. In this section we will look at two other important parts of circles: **chords** and **segments.**

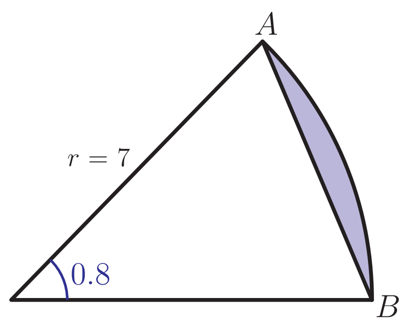


Worked example 11.17

The diagram shows a sector of a circle of radius 7 cm and the angle at the centre 0.8 radians. Find:

(a) the perimeter of the shaded region

(b) the area of the shaded region



The perimeter is made up of the arc AB and the chord [AB]

(a) $p = \text{arc} + \text{chord}$

The formula for the length of the arc is $l = r\theta$

$l = 7 \times 0.8 = 5.6$ cm

The chord [AB] is the third side of the triangle ABC. As we know the other two sides and the angle between them, we can use the cosine rule. Remember that the angle is in radians

Cosine rule:

$AB^2 = 7^2 + 7^2 - 2 \times 7 \times 7 \cos 0.8$

$AB^2 = 29.7$

$AB = \sqrt{29.7} = 5.45$ cm

continued . . .

We can now find the perimeter

$\therefore p = 5.6 + 5.45 = 11.1\,\text{cm}$

We know how to calculate the area of a sector. If we subtract the area of triangle *ABC*, we are left with the area of the segment

(b) $area = sector - triangle$

The formula for the area of a sector is $\frac{1}{2}r^2\theta$

$sector = \frac{1}{2}(7^2 \times 0.8) = 19.6\ \text{cm}^2$

The formula for the area of a triangle is $\frac{1}{2}ab\sin C$

$triangle = \frac{1}{2}(7 \times 7)\sin 0.8 = 17.58\,\text{cm}^2$

We can now find the area of the segment

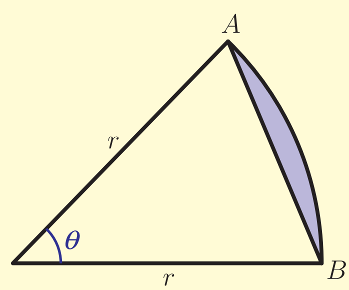
$\therefore area = 19.6 - 17.6 = 2.02\,\text{cm}^2$

We can follow the same method to derive general formulae for the length of a chord and area of a segment.

EXAM HINT

These formulae are not given in the Formula booklet, so you need to know how to derive them.

KEY POINT 11.10



The length of a chord of a circle is given by:

$$AB^2 = 2r^2(1 - \cos\theta)$$

and **the area of the shaded segment** is

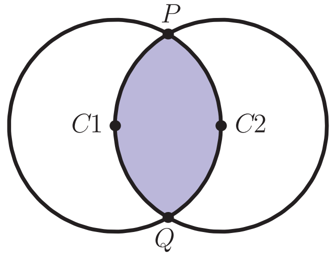
$$\frac{1}{2}r^2(\theta - \sin\theta),$$

where angle θ is measured in radians.

The next example shows how we can solve more complex geometry problems by splitting up the figure into basic shapes such as triangles and sectors.

Worked example 11.18

The diagram shows two equal circles of radius 12 such that the centre of one circle is on the circumference of the other.

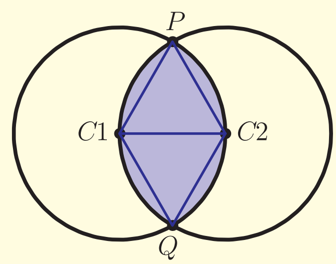


(a) Find the exact size of angle $P\hat{C}_1Q$ in radians.

(b) Calculate the exact area of the shaded region.

The only thing we know is the radius of the circle, so draw all the lengths which are equal to the radius

(a)



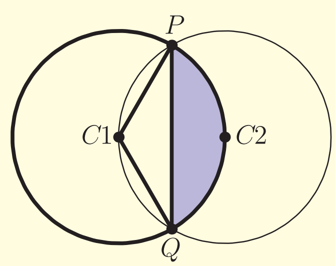
The lengths C_1P, C_2P and C_1C_2 are all equal to the radius of the circle. Therefore triangle PC_1C_2 is equilateral

$P\hat{C}_1C_2 = \frac{\pi}{3}$

$\therefore P\hat{C}_1Q = \frac{2\pi}{3}$

The shaded area is made up of two segments, each with angle at the centre $\frac{2\pi}{3}$

(b)



We can find area of one segment using the formula. Remember to use the exact value of $\sin\left(\frac{2\pi}{3}\right)$

area of one segment $= \frac{1}{2}(12^2)\left(\frac{2\pi}{3} - \sin\left(\frac{2\pi}{3}\right)\right)$

$= 72\left(\frac{2\pi}{3} - \frac{\sqrt{3}}{2}\right)$

$= 48\pi - 36\sqrt{3}$

The shaded area consists of two segments

$\therefore$ shaded area $= 96\pi - 72\sqrt{3}$

14 Lines and planes in space

In this chapter you will learn:

- to describe all points in space which lie on the same straight line
- to describe all points in space which lie in the same plane
- to solve three-dimensional problems involving intersections, distances and angles between lines and planes
- the connection between solving systems of linear equations and finding intersections of planes.

Introductory problem

Which is more stable (less wobbly): a three-legged stool or a four-legged stool?

Answering the above question involves thinking about whether given points lie in the same plane (on the same flat surface). In this chapter we will use the work on vectors from the last chapter to solve problems involving points, lines and planes in three-dimensional space. Such calculations are used in many areas such as design, navigation and computer games.

14A Vector equation of a line

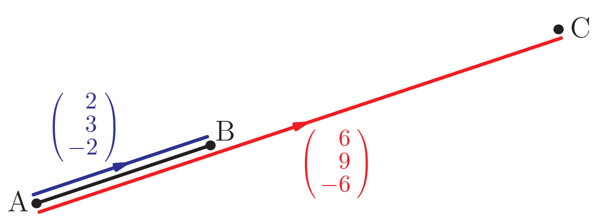
To solve problems involving lines in space we need a way of deciding whether a point lies on a given straight line.

Suppose we have two points, $A\ (-1, 1, 4)$ and $B\ (1, 4, 2)$. These two points determine a unique straight line (a 'straight line' extends indefinitely in both directions).

How can we check whether a third point lies on the same line?

We can use vectors to answer this question. For example, consider the point $C\ (5, 10, -2)$.

Then $\overrightarrow{AC} = \begin{pmatrix} 6 \\ 9 \\ -6 \end{pmatrix} = 3\begin{pmatrix} 2 \\ 3 \\ -2 \end{pmatrix} = 3\overrightarrow{AB}$.



EXAM HINT

Remember the IB notation for lines and line segments: (AB) means the (infinite) straight line through A and B, $[AB]$ is the line segment (the part of the line between A and B), and AB is the length of $[AB]$.

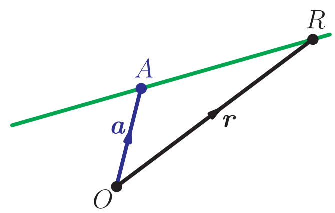
This means that (AC) is parallel to (AB). Since they both contain the point A, (AC) and (AB) must be the same straight line; in other words, C lies on the line (AB).

Next we ask how we can characterise all the points on the line (AB).

Using the above idea, we realise that a point R lies on (AB) if (AR) and (AB) are parallel. We know that this can be expressed using vectors by saying that $\overrightarrow{AR} = \lambda \overrightarrow{AB}$ for some value of the scalar λ (remember that a scalar means a number):

$$\overrightarrow{AR} = \begin{pmatrix} 2\lambda \\ 3\lambda \\ -2\lambda \end{pmatrix}.$$

We also know that $\overrightarrow{AR} = \boldsymbol{r} - \boldsymbol{a}$, where $\boldsymbol{r}$ and $\boldsymbol{a}$ are the position vectors of R and A, respectively.



See Section 13B for a reminder of vector algebra.

This means that $\boldsymbol{r} = \begin{pmatrix} -1 \\ 1 \\ 4 \end{pmatrix} + \begin{pmatrix} 2\lambda \\ 3\lambda \\ -2\lambda \end{pmatrix}$ is the position vector of a general point on the line (AB). In other words, R has coordinates $(-1+2\lambda, 1+3\lambda, 4-2\lambda)$ for some value of λ. Different values of λ correspond to different points on the line; for example, $\lambda = 0$ corresponds to point A, $\lambda = 1$ to point B and $\lambda = 3$ to point C.

You will see on page 416 that there is more than one possible vector equation of a line.

The line is parallel to the vector $\begin{pmatrix} 2 \\ 3 \\ -2 \end{pmatrix}$, so this vector determines the direction of the line. The expression for the position vector of $\boldsymbol{r}$ is usually written as $\boldsymbol{r} = \begin{pmatrix} -1 \\ 1 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 3 \\ -2 \end{pmatrix}$, so it is easy to identify the direction vector.

KEY POINT 14.1

The expression $\boldsymbol{r} = \boldsymbol{a} + \lambda\boldsymbol{d}$ is a **vector equation** of the line.

The vector $\boldsymbol{d}$ is the direction vector of the line and $\boldsymbol{a}$ is the position vector of one point on the line.

The vector $\boldsymbol{r}$ is the position vector of a general point on the line; different values of parameter λ give positions of different points on the line.

EXAM HINT

The formula booklet tells you the equation, but not what all the letters stand for.

Worked example 14.1

Write down a vector equation of the line passing through the point (–1, 1, 2) in the direction of the vector $\begin{pmatrix}2\\2\\1\end{pmatrix}$.

The equation of the line is $\boldsymbol{r} = \boldsymbol{a} + \lambda\boldsymbol{d}$, where $\boldsymbol{a}$ is the position vector of a point on the line and $\boldsymbol{d}$ is the direction vector

$$\underline{r} = \begin{pmatrix}-1\\1\\2\end{pmatrix} + \lambda\begin{pmatrix}2\\2\\1\end{pmatrix}$$

You know that, in two dimensions, a straight line is determined by its gradient and one point. The gradient is a value which determines the direction of the line. For example, for a line with gradient 3, an increase of 1 unit in x produces an increase of 3 units in y and so the line is in the direction of the vector $\begin{pmatrix}1\\3\end{pmatrix}$.

In three dimensions, a straight line is still determined by its direction and one point, but we cannot use a single number for the gradient. The line in Worked example 14.1 had a direction vector $\begin{pmatrix}2\\2\\1\end{pmatrix}$, so an increase of 2 units in x produces *both* an increase of 2 units in y *and* an increase of 1 unit in z.

Two points determine a straight line. The next example shows how to find a vector equation when two points on the line are given.

Worked example 14.2

Find a vector equation of the line through points A (–1, 1, 2) and B (3, 5, 4).

For a vector equation of the line, we need one point and the direction vector

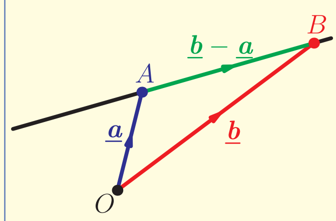
$$\underline{r} = \underline{a} + \lambda\underline{d}$$

The line passes through A (–1, 1, 2); we could use B (3, 5, 4) instead, but smaller values are preferable

$$\underline{a} = \begin{pmatrix}-1\\1\\2\end{pmatrix}$$

continued . . .

The line is in the direction of $\overrightarrow{AB}$, as we can see by drawing a diagram.



$\underline{d} = \overrightarrow{AB} = \underline{b} - \underline{a} = \begin{pmatrix} 4 \\ 4 \\ 2 \end{pmatrix}$

$\therefore \underline{r} = \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ 4 \\ 2 \end{pmatrix}$

What if we had used point B instead of A? Then we would have got the equation $\mathbf{r} = \begin{pmatrix} 3 \\ 5 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ 4 \\ 2 \end{pmatrix}$. This equation represents the same line, but the value of the parameter λ corresponding to any particular point will be different. For example, with the first equation point A has $\lambda = 0$ and point B has $\lambda = 1$, while using the second equation point A has $\lambda = -1$ and point B has $\lambda = 0$.

Note that the direction vector is not unique either, since we are only interested in its direction and not its magnitude.

Hence $\begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}$ or $\begin{pmatrix} -6 \\ -6 \\ -3 \end{pmatrix}$ could also be used as direction vectors for the above line, as they are all in the same direction. So yet another form of the equation of the same line would be

$\mathbf{r} = \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} -6 \\ -6 \\ -3 \end{pmatrix}$ and with this equation, point A has $\lambda = 0$ and point B has $\lambda = -\frac{2}{3}$.

To simplify calculations, we usually choose the direction vector to be the one involving smallest possible integer values, although sometimes it is more convenient to use the corresponding unit vector.

See Key point 13.3 about parallel vectors.

Worked example 14.3

(a) Show that the equations $r = \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}$ and $r = \begin{pmatrix} 5 \\ 7 \\ 5 \end{pmatrix} + \mu \begin{pmatrix} 6 \\ 6 \\ 3 \end{pmatrix}$ both represent the same straight line.

(b) Show that the equation $r = \begin{pmatrix} -5 \\ -3 \\ 1 \end{pmatrix} + t \begin{pmatrix} -4 \\ -4 \\ -2 \end{pmatrix}$ represents a different straight line.

EXAM HINT

When a question asks for equations of several lines, then different letters should be used for the parameters. Commonly used letters are λ (lambda), μ (mu), *t* and *s*.

We need to show that the two lines have parallel direction vectors (they will then be parallel) and one common point (then they will be the same line)

Two vectors are parallel if one is a scalar multiple of the other

(a) Direction vectors are parallel, as

$$\begin{pmatrix} 6 \\ 6 \\ 3 \end{pmatrix} = 3 \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}$$

The point (5, 7, 5) will lie on the first line if we can find the value of λ which gives this position vector

Show that (5, 7, 5) lies on the first line:

Find the value of λ which gives the first coordinate

$$-1 + 2\lambda = 5$$
$$\therefore \lambda = 3$$

This value of λ must give the other two coordinates

$$\begin{cases} 1 + 3 \times 2 = 7 \\ 2 + 3 \times 1 = 5 \end{cases}$$

∴ (5, 7, 5) lies on the line.

Hence the two lines are the same.

Check whether the direction vectors are parallel

(b) $\begin{pmatrix} -4 \\ -4 \\ -2 \end{pmatrix} = -2 \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}$

So the line is parallel to the other two.

Check whether (5, –3,1) lies on the first line. Find the value of λ which gives the first coordinate

$$-1 + 2\lambda = -5$$
$$\therefore \lambda = -2$$

continued . . .

This value of λ must give the other two coordinates

$$\begin{cases} 1+(-2)\times 2=-3 \\ 2+(-2)\times 1=0\neq 1 \end{cases}$$

(5, −3, 1) *does not lie on the line.*

Hence the line is not the same as the first line.

In the above example we used the coordinates of the point to find the corresponding value of λ. However, sometimes we only know that a point lies on the line, but not its precise coordinates. The next example shows how we can work with a general point on the line (with an unknown value of λ).

Worked example 14.4

Point B (3, 5, 4) lies on the line with equation $r = \begin{pmatrix}-1\\1\\2\end{pmatrix}+\lambda\begin{pmatrix}2\\2\\1\end{pmatrix}$. Find the possible positions of a point Q on the line such that $BQ = 15$.

We know that Q lies on the line, so it has the position vector $\begin{pmatrix}-1\\1\\2\end{pmatrix}+\lambda\begin{pmatrix}2\\2\\1\end{pmatrix}$ for some value of λ, and we need to find λ.

$$\underline{q}=\begin{pmatrix}-1\\1\\2\end{pmatrix}+\lambda\begin{pmatrix}2\\2\\1\end{pmatrix}=\begin{pmatrix}-1+2\lambda\\1+2\lambda\\2+\lambda\end{pmatrix}$$

We can express vector $\overrightarrow{BQ}$ in terms of λ and then set its magnitude equal to 15

$$\overrightarrow{BQ}=\underline{q}-\underline{b}$$

$$=\begin{pmatrix}-1+2\lambda\\1+2\lambda\\2+\lambda\end{pmatrix}-\begin{pmatrix}3\\5\\4\end{pmatrix}=\begin{pmatrix}2\lambda-4\\2\lambda-4\\\lambda-2\end{pmatrix}$$

$$\left|\overrightarrow{BQ}\right|=15$$

It is easier to work without the square root, so we square the magnitude equation

$$\therefore (2\lambda-4)^2+(2\lambda-4)^2+(\lambda-2)^2=15^2$$

$$\Leftrightarrow \quad 9\lambda^2-36\lambda-189=0$$

$$\Leftrightarrow \quad \lambda=-3 \text{ or } 7$$

We can now find the position vector of Q by substituting the values of λ in to the line equation

$$\therefore \underline{q}=\begin{pmatrix}-7\\-5\\-1\end{pmatrix} \text{ or } \begin{pmatrix}13\\15\\9\end{pmatrix}$$

Exercise 14A

1. Find the vector equation of each line in the given direction through the given point.

   (a) (i) Direction $\begin{pmatrix}1\\4\end{pmatrix}$, point $(4,-1)$

   (ii) Direction $\begin{pmatrix}2\\-3\end{pmatrix}$, point $(4,1)$

   (b) (i) Point $(1,0,5)$, direction $\begin{pmatrix}1\\3\\-3\end{pmatrix}$

   (ii) Point $(-1,1,5)$, direction $\begin{pmatrix}3\\-2\\2\end{pmatrix}$

   (c) (i) Point $(4,0)$, direction $2\boldsymbol{i}+3\boldsymbol{j}$

   (ii) Point $(0,2)$, direction $\boldsymbol{i}-3\boldsymbol{j}$

   (d) (i) Direction $\boldsymbol{i}-3\boldsymbol{k}$, point $(0,2,3)$

   (ii) Direction $2\boldsymbol{i}+3\boldsymbol{j}-\boldsymbol{k}$, point $(4,-3,0)$

2. Find a vector equation of the line through each pair of points.

   (a) (i) $(4,1)$ and $(1,2)$ (ii) $(2,7)$ and $(4,-2)$

   (b) (i) $(-5,-2,3)$ and $(4,-2,3)$ (ii) $(1,1,3)$ and $(10,-5,0)$

3. Decide whether or not the given point lies on the given line.

   (a) (i) Line $\boldsymbol{r}=\begin{pmatrix}2\\1\\5\end{pmatrix}+t\begin{pmatrix}-1\\2\\2\end{pmatrix}$, point $(0,\ 5,\ 9)$

   (ii) Line $\boldsymbol{r}=\begin{pmatrix}-1\\0\\3\end{pmatrix}+t\begin{pmatrix}4\\1\\5\end{pmatrix}$, point $(-1,\ 0,\ 3)$

   (b) (i) Line $\boldsymbol{r}=\begin{pmatrix}4\\0\\3\end{pmatrix}+t\begin{pmatrix}4\\0\\3\end{pmatrix}$, point $(0,\ 0,\ 0)$

   (ii) Line $\boldsymbol{r}=\begin{pmatrix}-1\\5\\1\end{pmatrix}+t\begin{pmatrix}0\\0\\7\end{pmatrix}$, point $(-1,\ 3,\ 8)$

4. (a) Show that the points A (4, −1, −8) and B (2, 1, −4) lie on the line l with equation $r = \begin{pmatrix} 2 \\ 1 \\ -4 \end{pmatrix} + t\begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}$.

(b) Find the coordinates of the point C on the line l such that $AB = BC$. *[6 marks]*

5. (a) Find the vector equation of line l through points P (7, 1, 2) and Q (3, −1, 5).

(b) Point R lies on l and $PR = 3PQ$. Find the possible coordinates of R. *[6 marks]*

6. (a) Write down the vector equation of the line l through the point A (2, 1, 4) parallel to the vector $2\boldsymbol{i} - 3\boldsymbol{j} + 6\boldsymbol{k}$.

(b) Calculate the magnitude of the vector $2\boldsymbol{i} - 3\boldsymbol{j} + 6\boldsymbol{k}$.

(c) Find the possible coordinates of point P on l such that $AP = 35$. *[8 marks]*

14B Solving problems with lines

In this section we will use vector equations of lines to solve problems involving angles and intersections.

We will also see how vector equations of lines can be used to describe paths of moving objects in mechanics.

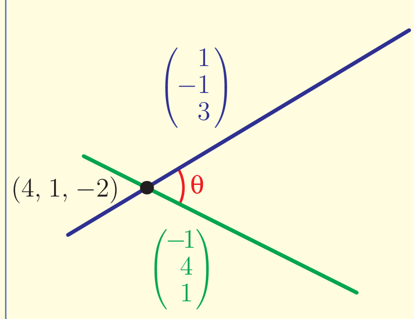
Worked example 14.5

Find the acute angle between lines with equations $\boldsymbol{r} = \begin{pmatrix} 4 \\ 1 \\ -2 \end{pmatrix} + t\begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix}$ and $\boldsymbol{r} = \begin{pmatrix} 4 \\ 1 \\ -2 \end{pmatrix} + \lambda\begin{pmatrix} -1 \\ 4 \\ 1 \end{pmatrix}$.

We know the formula for the angle between two vectors (see Section 13D)

$$\cos\theta = \frac{\boldsymbol{a}\bullet\boldsymbol{b}}{|\boldsymbol{a}||\boldsymbol{b}|}$$

Draw a diagram to identify which two vectors $\boldsymbol{a}$ and $\boldsymbol{b}$ make the required angle



continued . . .

The two vectors are in the directions of the two lines. So we take $\boldsymbol{a}$ and $\boldsymbol{b}$ to be the direction vectors of the two lines

$$\underline{a} = \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix}$$

$$\underline{b} = \begin{pmatrix} -1 \\ 4 \\ 1 \end{pmatrix}$$

We can now use the formula to calculate the angle

$$\therefore \quad \cos\theta = \frac{-1-4+3}{\sqrt{(1+1+9)}\sqrt{1+16+1}}$$

$$= -\frac{2}{\sqrt{11}\sqrt{18}}$$

$$\theta = 98.2°$$

The angle found is obtuse; the question asked for the acute angle

acute angle $= 180° - 98.2° = 81.8°$

The example above illustrates the general method for finding an angle between two lines.

KEY POINT 14.2

The angle between two lines is equal to the angle between their direction vectors.

Now that we know that the angle between two lines is the angle between their direction vectors, it is easy to identify parallel and perpendicular lines.

KEY POINT 14.3

Two lines with direction vectors $\boldsymbol{d}_1$ and $\boldsymbol{d}_2$ are:

- parallel if $\boldsymbol{d}_1 = k\,\boldsymbol{d}_2$
- perpendicular if $\boldsymbol{d}_1 \cdot \boldsymbol{d}_2 = 0$.

Parallel and perpendicular vectors were covered in Sections 13B and 13E.

Worked example 14.6

Decide whether the following pairs of lines are parallel, perpendicular, or neither:

(a) $\boldsymbol{r}=\begin{pmatrix}2\\-1\\5\end{pmatrix}+\lambda\begin{pmatrix}4\\-1\\2\end{pmatrix}$ and $\boldsymbol{r}=\begin{pmatrix}-2\\0\\3\end{pmatrix}+\mu\begin{pmatrix}1\\-2\\-3\end{pmatrix}$

(b) $\boldsymbol{r}=\begin{pmatrix}0\\0\\1\end{pmatrix}+\lambda\begin{pmatrix}2\\1\\1\end{pmatrix}$ and $\boldsymbol{r}=\begin{pmatrix}2\\1\\2\end{pmatrix}+t\begin{pmatrix}1\\0\\-3\end{pmatrix}$

(c) $\boldsymbol{r}=\begin{pmatrix}2\\-1\\5\end{pmatrix}+t\begin{pmatrix}4\\-6\\2\end{pmatrix}$ and $\boldsymbol{r}=\begin{pmatrix}-2\\0\\3\end{pmatrix}+s\begin{pmatrix}-10\\15\\-5\end{pmatrix}$

Is $\boldsymbol{d}_1$ a multiple of $\boldsymbol{d}_2$?

(a) If $\begin{pmatrix}4\\-1\\2\end{pmatrix}=k\begin{pmatrix}1\\-2\\-3\end{pmatrix}$ then

$$\begin{cases}4=k\times1\Rightarrow k=4\\-1=k\times(-2)\Rightarrow k=\dfrac{1}{2}\end{cases}$$

$4\neq\dfrac{1}{2}$

$\therefore$ They are not parallel.

Is $\boldsymbol{d}_1\cdot\boldsymbol{d}_2=0$?

$$\begin{pmatrix}4\\-1\\2\end{pmatrix}\cdot\begin{pmatrix}1\\-2\\-3\end{pmatrix}=4+2-6=0$$

$\therefore$ The lines are perpendicular.

Is $\boldsymbol{d}_1$ a multiple of $\boldsymbol{d}_2$?

(b) If $\begin{pmatrix}2\\1\\1\end{pmatrix}=k\begin{pmatrix}1\\0\\-3\end{pmatrix}$ then

$$\begin{cases}2=k\times1\Rightarrow k=2\\1=k\times0 \text{ impossible}\end{cases}$$

$\therefore$ They are not parallel.

Is $\boldsymbol{d}_1\cdot\boldsymbol{d}_2=0$?

$$\begin{pmatrix}2\\1\\2\end{pmatrix}\cdot\begin{pmatrix}1\\0\\3\end{pmatrix}=2+0+6=8\neq0$$

$\therefore$ The lines are neither parallel nor perpendicular.

continued . . .

Is $\boldsymbol{d}_1$ a multiple of $\boldsymbol{d}_2$?

(c) If $\begin{pmatrix} 4 \\ -6 \\ 2 \end{pmatrix} = k\begin{pmatrix} -10 \\ 15 \\ -5 \end{pmatrix}$ then

$$\begin{cases} 4 = k \times (-10) \Rightarrow k = -\frac{2}{5} \\ -6 = k \times 15 \Rightarrow k = -\frac{2}{5} \\ 2 = k \times (-5) \Rightarrow k = -\frac{2}{5} \end{cases}$$

∴ The lines have parallel directions.

Check to see if they are the same line

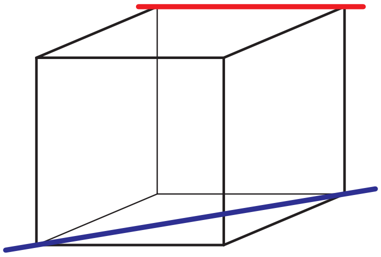
Point $\begin{pmatrix} -2 \\ 0 \\ 3 \end{pmatrix}$ does not lie on the first line:

$$\begin{cases} 2 + 4t = -2 \Rightarrow t = -1 \\ -1 - 6t = 0 \Rightarrow t = -\frac{1}{6} \end{cases}$$

They are not the same line.

∴ The lines are parallel.

We will now see how to find the point of intersection of two lines. Suppose two lines have vector equations $\boldsymbol{r}_1 = \boldsymbol{a} + \lambda \boldsymbol{d}_1$ and $\boldsymbol{r}_2 = \boldsymbol{b} + \mu \boldsymbol{d}_2$. If they intersect, then there must be a point which lies on both lines. Remembering that the position vector of a point on the line is given by the vector $\boldsymbol{r}$, this means that we need to find the values of λ and μ which make $\boldsymbol{r}_1 = \boldsymbol{r}_2$.



In two dimensions, two straight lines either intersect or are parallel. However, in three dimensions it is possible to have two lines which are not parallel but do not intersect, as illustrated by the red and blue lines in the diagram. Such lines are called **skew lines**.

With skew lines we will see that we cannot find values of λ and μ such that $\boldsymbol{r}_1 = \boldsymbol{r}_2$.

Worked example 14.7

Find the coordinates of the point of intersection of the following pairs of lines.

(a) $\boldsymbol{r}=\begin{pmatrix}0\\-4\\1\end{pmatrix}+\lambda\begin{pmatrix}1\\2\\1\end{pmatrix}$ and $\boldsymbol{r}=\begin{pmatrix}1\\3\\5\end{pmatrix}+\mu\begin{pmatrix}4\\-2\\-2\end{pmatrix}$ (b) $\boldsymbol{r}=\begin{pmatrix}-4\\1\\3\end{pmatrix}+t\begin{pmatrix}1\\1\\4\end{pmatrix}$ and $\boldsymbol{r}=\begin{pmatrix}2\\1\\1\end{pmatrix}+\lambda\begin{pmatrix}2\\-3\\2\end{pmatrix}$

We need to make $\boldsymbol{r}_1=\boldsymbol{r}_2$

(a) $\begin{pmatrix}0\\-4\\1\end{pmatrix}+\lambda\begin{pmatrix}1\\2\\1\end{pmatrix}=\begin{pmatrix}1\\3\\5\end{pmatrix}+\mu\begin{pmatrix}4\\-2\\-2\end{pmatrix}$

$$\Leftrightarrow\begin{pmatrix}0+\lambda\\-4+2\lambda\\1+\lambda\end{pmatrix}=\begin{pmatrix}1+4\mu\\3-2\mu\\5-2\mu\end{pmatrix}$$

If two vectors are equal, then all their components are equal

$$\Rightarrow\begin{cases}0+\lambda=1+4\mu\\-4+2\lambda=3-2\mu\\1+\lambda=5-2\mu\end{cases}$$

$$\Rightarrow\begin{cases}\lambda-4\mu=1 & (1)\\2\lambda+2\mu=7 & (2)\\\lambda+2\mu=4 & (3)\end{cases}$$

Solve two simultaneous equations in two variables. Use eqn (1) and (3) (as subtracting them eliminates λ)

$(3)-(1)\quad 6\mu=3$

$\therefore \mu=\frac{1}{2}, \lambda=3$

We need to check that the values of λ and μ also satisfy the second equation otherwise the lines do not actually meet

$(2): 2\times3+2\times\frac{1}{2}=7$

$\therefore$ the lines intersect

The position of the intersection point is given by the vector $\boldsymbol{r}_1$ (or $\boldsymbol{r}_2$ - they should be the same)

$$\underline{r}_1=\begin{pmatrix}0\\-4\\1\end{pmatrix}+3\begin{pmatrix}1\\2\\1\end{pmatrix}=\begin{pmatrix}3\\2\\4\end{pmatrix}$$

The lines intersect at the point $(3, 2, 4)$

Make $\boldsymbol{r}_1=\boldsymbol{r}_2$

(b) $\begin{pmatrix}-4\\3\\3\end{pmatrix}+t\begin{pmatrix}1\\1\\4\end{pmatrix}=\begin{pmatrix}2\\1\\1\end{pmatrix}+\lambda\begin{pmatrix}2\\-3\\2\end{pmatrix}$

$$\Rightarrow\begin{cases}t-2\lambda=6 & (1)\\t+3\lambda=-2 & (2)\\4t-2\lambda=-2 & (3)\end{cases}$$

continued . . .

We can find t and λ from eqn (1) and (2)

$$(1) \text{ and } (2) \Rightarrow \lambda = -\frac{8}{5}, t = \frac{14}{5}$$

We need to check that the values found also satisfy the third equation

$$(3) \quad 4 \times \frac{14}{5} - 2 \times \left(-\frac{8}{5}\right) = \frac{72}{5} \neq -2$$

This tells us that it is impossible to find t and λ to make $\boldsymbol{r}_1 = \boldsymbol{r}_2$

The two lines do not intersect.

EXAM HINT

You can use your calculator to solve simultaneous equations.
See Calculator sheet 6 on the CD-ROM.

Vector questions often ask you to find a point on a given line which satisfies certain conditions. We have already seen how we can use the position vector $\boldsymbol{r}$ for a general point on the line, and then use the condition to write an equation for λ.

See Worked example 14.4.

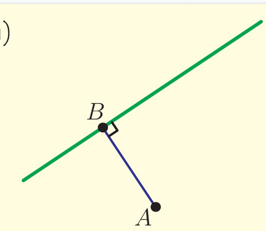
Worked example 14.8

Line l has equation $\boldsymbol{r} = \begin{pmatrix} 3 \\ -1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$ and point A has coordinates (3, 9, –2).

(a) Find the coordinates of point B on l so that AB is perpendicular to l.
(b) Hence find the shortest distance from A to l.
(c) Find the coordinates of the reflection of the point A in l.

Draw a diagram. The line AB should be perpendicular to the direction vector of l

(a)



$$\overrightarrow{AB} \cdot \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} = 0 \qquad (1)$$

continued . . .

We know that B lies on l, so its position vector is given by the equation for $\boldsymbol{r}$

$$\overrightarrow{OB} = \boldsymbol{r} = \begin{pmatrix} 3+\lambda \\ -1-\lambda \\ \lambda \end{pmatrix}$$

$\overrightarrow{AB} = \boldsymbol{b} - \boldsymbol{a}$

$$\therefore \overrightarrow{AB} = \begin{pmatrix} 3+\lambda \\ -1-\lambda \\ \lambda \end{pmatrix} - \begin{pmatrix} 3 \\ 9 \\ -2 \end{pmatrix} = \begin{pmatrix} \lambda \\ -10-\lambda \\ \lambda+2 \end{pmatrix}$$

Now find the value of λ for which the two lines are perpendicular

$$\begin{pmatrix} \lambda \\ -10-\lambda \\ \lambda+2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} = 0$$

$$\Rightarrow (\lambda) + (10+\lambda) + (\lambda+2) = 0$$

$$\Rightarrow \lambda = -4$$

Use value of λ in the equation of the line to give the position vector of B

$$r = \begin{pmatrix} -1 \\ 3 \\ -4 \end{pmatrix}$$

$\therefore$ B has coordinates $(-1, 3, -4)$

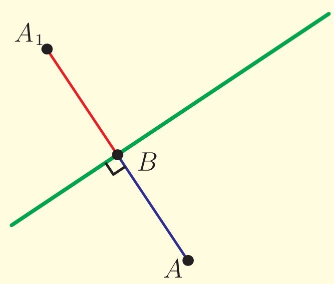
The shortest distance from a point to a line is the perpendicular distance AB. Again, use $\overrightarrow{AB} = \boldsymbol{b} - \boldsymbol{a}$

(b) $$\overrightarrow{AB} = \begin{pmatrix} -1 \\ 3 \\ -4 \end{pmatrix} - \begin{pmatrix} 3 \\ 9 \\ -2 \end{pmatrix} = \begin{pmatrix} -4 \\ -6 \\ -2 \end{pmatrix}$$

$$\therefore \left|\overrightarrow{AB}\right| = \sqrt{16+36+4} = 2\sqrt{14}$$

The reflection A_1 lies on the line (AB). Since $BA_1 = AB$ and they are also in the same direction, $\overrightarrow{BA_1} = \overrightarrow{AB}$

(c)



$$\overrightarrow{BA_1} = \overrightarrow{AB}$$

$$\Rightarrow \underline{a}_1 - \underline{b} = \overrightarrow{AB}$$

$$\therefore \underline{a}_1 = \begin{pmatrix} -4 \\ -6 \\ -2 \end{pmatrix} + \begin{pmatrix} -1 \\ 3 \\ -4 \end{pmatrix}$$

So A_1 has coordinates $(-5, -3, -6)$

Worked example 14.8(c) illustrates the power of vectors. As vectors contain both distance and direction information, just one equation ($\overrightarrow{BA_1} = \overrightarrow{AB}$) was needed to express both the fact that A_1 lies on the line (AB) and that $BA_1 = AB$.

We have already mentioned that vectors have many applications, particularly in physics. One such application is describing positions, displacements and velocities. These are all vector quantities, since they have both magnitude and direction.

You are probably familiar with the rule that, for an object moving with constant velocity, displacement = velocity × time. If we are working in two or three dimensions, the positions of points also need to be described by vectors. Suppose an object has constant velocity $\boldsymbol{v}$ and in time t moves from the point with position $\boldsymbol{a}$ to the point with position $\boldsymbol{r}$. Then its displacement is $\boldsymbol{r}-\boldsymbol{a}$, so we can write:

$$\boldsymbol{r}-\boldsymbol{a}=\boldsymbol{v}t$$

This equation can be rearranged to $\boldsymbol{r}=\boldsymbol{a}+t\boldsymbol{v}$, which looks very much like a vector equation of a line with direction vector $\boldsymbol{v}$. This makes sense, as the object will move in the direction given by its velocity vector. As t changes, $\boldsymbol{r}$ gives position vectors of different points along the object's path.

Note that the speed is the magnitude of the velocity, $|\boldsymbol{v}|$, and the distance travelled is the magnitude of the displacement, $|\boldsymbol{r}-\boldsymbol{a}|$.

In 1896 the physicist Lord Kelvin wrote: "'vector' is a useless survival, or offshoot from quaternions, and has never been of the slightest use to any creature".

(Quaternions are a special type of number linked to complex numbers.)

They are now one of the most important tools in physics. Even great mathematicians cannot always predict what will be useful!

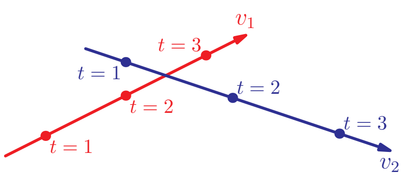
KEY POINT 14.4

For an object moving with constant velocity $\boldsymbol{v}$ from an initial position $\boldsymbol{a}$, the position at time t is given by
$\boldsymbol{r}(t)=\boldsymbol{a}+t\boldsymbol{v}$.

The object moves along the straight line with equation $\boldsymbol{r}=\boldsymbol{a}+t\boldsymbol{v}$.

The speed of the object is equal to $|\boldsymbol{v}|$.

When we wanted to find the intersection of two lines, we had to use different parameters (for example, λ and μ) in the two equations. If we have two objects, we can write an equation for $\boldsymbol{r}(t)$ for each of them. In this case, we should use the same t in both equations, as both objects are moving at the same time. For the two objects to meet, they need to be at the same place at the same time. Notice that it is possible for the objects' paths to cross without the objects themselves meeting, if they pass through the intersection point at different times.



Worked example 14.9

Two objects, A and B, have velocities $\boldsymbol{v}_A = 6\boldsymbol{i} + 3\boldsymbol{j} + \boldsymbol{k}$ and $\boldsymbol{v}_B = -2\boldsymbol{i} + \boldsymbol{j} + 7\boldsymbol{k}$. Object A starts from the origin and object B from the point with position vector $13\boldsymbol{i} - \boldsymbol{j} + 3\boldsymbol{k}$. Distance is measured in kilometres and time in hours.

(a) What is the speed of object B?

(b) Find the distance between the two objects after 5 hours.

(c) Show that the two objects do not meet.

Speed is the magnitude of velocity

(a)

$$|v_B| = \sqrt{2^2 + 1^2 + 7^2} = \sqrt{54}$$

So the speed of B is 7.35 km/h.

We need an equation for the position of each object in terms of t

(b)

Using $\boldsymbol{r}(t) = \boldsymbol{a} + t\boldsymbol{v}$:

$$\boldsymbol{r}_A(t) = t(6\boldsymbol{i} + 3\boldsymbol{j} + \boldsymbol{k})$$

$$\boldsymbol{r}_B(t) = 13\boldsymbol{i} - \boldsymbol{j} + 3\boldsymbol{k} + t(-2\boldsymbol{i} + \boldsymbol{j} + 7\boldsymbol{k})$$

We can then find the position of each object when $t = 5$

When $t = 5$:

$$\boldsymbol{r}_A = 30\boldsymbol{i} + 15\boldsymbol{j} + 5\boldsymbol{k}$$

$$\boldsymbol{r}_B = 3\boldsymbol{i} + 4\boldsymbol{j} + 38\boldsymbol{k}$$

The distance is the magnitude of $\boldsymbol{r}_A - \boldsymbol{r}_B$

$$|\boldsymbol{r}_A - \boldsymbol{r}_B| = \sqrt{27^2 + 11^2 + 33^2}$$

$$= 44.0\text{ km}$$

If the two objects meet then $\boldsymbol{r}_A(t) = \boldsymbol{r}_B(t)$

(c)

If $\boldsymbol{r}_A(t) = \boldsymbol{r}_B(t)$:

$$\begin{cases} 6t = 13 - 2t \Rightarrow t = \frac{13}{8} \\ 3t = -1 + t \Rightarrow t = -\frac{1}{2} \\ t = 3 + 7t \Rightarrow t = -\frac{1}{2} \end{cases}$$

The three coordinates are not equal at the same time, so the objects do not meet.

Exercise 14B

1. Find the acute angle between the following pairs of lines, giving your answer in degrees.

(a) (i) $\boldsymbol{r}=\begin{pmatrix}5\\-1\\2\end{pmatrix}+\lambda\begin{pmatrix}2\\2\\3\end{pmatrix}$ and $\boldsymbol{r}=\begin{pmatrix}1\\1\\0\end{pmatrix}+\mu\begin{pmatrix}4\\-1\\3\end{pmatrix}$

(ii) $\boldsymbol{r}=\begin{pmatrix}4\\0\\2\end{pmatrix}+\lambda\begin{pmatrix}2\\-1\\1\end{pmatrix}$ and $\boldsymbol{r}=\begin{pmatrix}1\\0\\2\end{pmatrix}+\mu\begin{pmatrix}-5\\1\\3\end{pmatrix}$

(b) (i) $\boldsymbol{r}=\begin{pmatrix}2\\0\\1\end{pmatrix}+t\begin{pmatrix}-1\\0\\0\end{pmatrix}$ and $\boldsymbol{r}=\begin{pmatrix}1\\3\\3\end{pmatrix}+s\begin{pmatrix}4\\0\\2\end{pmatrix}$

(ii) $\boldsymbol{r}=\begin{pmatrix}6\\6\\2\end{pmatrix}+t\begin{pmatrix}-1\\0\\3\end{pmatrix}$ and $\boldsymbol{r}=\begin{pmatrix}1\\0\\0\end{pmatrix}+s\begin{pmatrix}4\\-1\\2\end{pmatrix}$

2. For each pair of lines, state whether they are parallel, perpendicular, the same line, or none of the above.

(a) $\boldsymbol{r}=\begin{pmatrix}0\\0\\1\end{pmatrix}+\text{»}\begin{pmatrix}2\\-1\\3\end{pmatrix}$ and $\boldsymbol{r}=\begin{pmatrix}0\\0\\1\end{pmatrix}+\mu\begin{pmatrix}2\\1\\-1\end{pmatrix}$

(b) $\boldsymbol{r}=\begin{pmatrix}4\\1\\2\end{pmatrix}+s\begin{pmatrix}-1\\2\\2\end{pmatrix}$ and $\boldsymbol{r}=\begin{pmatrix}2\\1\\1\end{pmatrix}+t\begin{pmatrix}2\\-4\\-4\end{pmatrix}$

(c) $\boldsymbol{r}=\begin{pmatrix}1\\5\\2\end{pmatrix}+\lambda\begin{pmatrix}3\\3\\1\end{pmatrix}$ and $\boldsymbol{r}=\begin{pmatrix}0\\0\\1\end{pmatrix}+t\begin{pmatrix}3\\3\\1\end{pmatrix}$

(d) $\boldsymbol{r}=\begin{pmatrix}2\\2\\1\end{pmatrix}+t\begin{pmatrix}1\\-1\\3\end{pmatrix}$ and $\boldsymbol{r}=\begin{pmatrix}5\\-1\\10\end{pmatrix}+s\begin{pmatrix}1\\-1\\3\end{pmatrix}$

3. Determine whether the following pairs of lines intersect and, if they do, find the coordinates of the intersection point.

(a) (i) $\boldsymbol{r}=\begin{pmatrix}6\\1\\2\end{pmatrix}+\lambda\begin{pmatrix}-1\\2\\1\end{pmatrix}$ and $\boldsymbol{r}=\begin{pmatrix}2\\1\\-14\end{pmatrix}+\mu\begin{pmatrix}2\\-2\\3\end{pmatrix}$

(ii) $\boldsymbol{r}=\begin{pmatrix}4\\-1\\2\end{pmatrix}+\lambda\begin{pmatrix}1\\2\\-4\end{pmatrix}$ and $\boldsymbol{r}=\begin{pmatrix}6\\-2\\0\end{pmatrix}+\mu\begin{pmatrix}3\\-4\\0\end{pmatrix}$

(b) (i) $\boldsymbol{r}=\begin{pmatrix}1\\2\\3\end{pmatrix}+t\begin{pmatrix}-1\\1\\2\end{pmatrix}$ and $\boldsymbol{r}=\begin{pmatrix}-4\\-4\\-11\end{pmatrix}+s\begin{pmatrix}5\\1\\2\end{pmatrix}$

(ii) $\boldsymbol{r}=\begin{pmatrix}4\\0\\2\end{pmatrix}+t\begin{pmatrix}2\\0\\1\end{pmatrix}$ and $\boldsymbol{r}=\begin{pmatrix}-1\\2\\3\end{pmatrix}+s\begin{pmatrix}1\\-2\\-2\end{pmatrix}$

4. Line l has equation $\boldsymbol{r}=\begin{pmatrix}4\\2\\-1\end{pmatrix}+\lambda\begin{pmatrix}2\\-1\\2\end{pmatrix}$ and point P has coordinates (7, 2, 3).

Point C lies on l and PC is perpendicular to l. Find the coordinates of C. *[6 marks]*

5. Find the shortest distance from the point (−1, 1, 2) to the line with equation $\boldsymbol{r}=\begin{pmatrix}1\\0\\2\end{pmatrix}+t\begin{pmatrix}-3\\1\\1\end{pmatrix}$. *[6 marks]*

6. Two lines are given by $l_1:\boldsymbol{r}=\begin{pmatrix}-5\\1\\10\end{pmatrix}+\lambda\begin{pmatrix}-3\\0\\4\end{pmatrix}$ and

$l_2:\boldsymbol{r}=\begin{pmatrix}3\\0\\-9\end{pmatrix}+\mu\begin{pmatrix}1\\1\\7\end{pmatrix}$.

(a) l_1 and l_2 intersect at P, find the coordinates of P.

(b) Show that the point Q (5, 2, 5) lies on l_2.

(c) Find the coordinates of point M on l_1 such that QM is perpendicular to l_1.

(d) Find the area of the triangle PQM. *[10 marks]*

7. In this question, unit vectors $\boldsymbol{i}$ and $\boldsymbol{j}$ point due East and North, respectively.

A port is located at the origin. One ship starts from the port and moves with velocity $\boldsymbol{v}_1 = (3\boldsymbol{i} + 4\boldsymbol{j})$ kmh^{-1}.

(a) Write down the position vector at time t hours.

At the same time, a second ship starts 18 km north of the port and moves with velocity $\boldsymbol{v}_2 = (3\boldsymbol{i} - 5\boldsymbol{j})$ kmh^{-1}.

(b) Write down the position vector of the second ship at time t hours.

(c) Show that after half an hour, the distance between the two ships is 13.5 km.

(d) Show that the ships meet, and find the time when this happens.

(e) How long after the meeting are the ships 18 km apart?

[12 marks]

8. At time $t = 0$, two aircraft have position vectors $5\boldsymbol{j}$ and $7\boldsymbol{k}$. The first moves with velocity $3\boldsymbol{i} - 4\boldsymbol{j} + \boldsymbol{k}$ and the second with velocity $5\boldsymbol{i} + 2\boldsymbol{j} - \boldsymbol{k}$.

(a) Write down the position vector of the first aircraft at time t.

(b) Show that at time t the distance, d, between the two aircraft is given by $d^2 = 44t^2 - 88t + 74$.

(c) Show that the two aircraft will not collide.

(d) Find the minimum distance between the two aircraft.

[12 marks]

9. Find the distance of the line with equation $\boldsymbol{r} = \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}$ from the origin.

[7 marks]

10. Two lines with equations $l_1 : \boldsymbol{r} = \begin{pmatrix} 0 \\ -1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 5 \\ 3 \end{pmatrix}$ and $l_2 : \boldsymbol{r} = \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix} + t \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix}$ intersect at point P.

(a) Find the coordinates of P.

(b) Find, in degrees, the acute angle between the two lines.

Point Q has coordinates $(-1, 5, 10)$.

(c) Show that Q lies on l_2.

(d) Find the distance PQ.

(e) Hence find the shortest distance from Q to the line l_1. *[12 marks]*

11. Given line $l: \boldsymbol{r} = \begin{pmatrix} 5 \\ 1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -3 \\ 3 \end{pmatrix}$ and point P (21, 5, 10):

(a) Find the coordinates of point M on l such that PM is perpendicular to l.

(b) Show that the point Q (15, −14, 17) lies on l.

(c) Find the coordinates of point R on l_1 such that $|PR| = |PQ|$. *[10 marks]*

12. Two lines have equations $l_1: \boldsymbol{r} = \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix}$ and

$l_2: \boldsymbol{r} = \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$ and intersect at point P.

(a) Show that Q (5, 2, 6) lies on l_2.

(b) R is a point on l_1 such that $|PR| = |PQ|$. Find the possible coordinates of R. *[8 marks]*

14C Other forms of equation of a line

You know that in two dimensions, a straight line has equation of the form $y = mx + c$ or $ax + by = c$. How is this related to the vector equation of the line we introduced in this chapter?

Let us look at an example of a vector equation of a line in two dimensions. A line with direction vector $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$ passing through the point (1, 4) has vector equation $\boldsymbol{r} = \begin{pmatrix} 1 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 2 \end{pmatrix}$. Vector $\boldsymbol{r}$ is the position vector of a point on the line; in other words, it gives